The pair singular series of a polynomial III: the off-diagonal in Cesàro form

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Abstract

For an irreducible quadratic f with pair singular series $S_f(h)$ and Bateman–Horn constant $C(f)$, parts I and II reduced the conjecture $\sum_{h\le H}(S_f(h)-C(f)^2) = -\frac{1}{2}C(f)\log H + A_f + o(1)$ to a statement about the pairs of distinct roots of f modulo squarefree d ("Hypothesis (E)"), and showed that in Cesàro form any unbounded saving over the trivial bound suffices. This paper takes Hypothesis (E) apart. We prove an exact decomposition of the Cesàro off-diagonal into pieces indexed by the part u of the modulus on which the two roots agree: piece u is the same problem with length H/u and with the roots of $x^2 \equiv D$ replaced by the roots of $u^2x^2 \equiv D$, a dilation by u^{-1}, and all pieces carry equal weight per unit of $\log u$. Three parts are unconditional: the piece $u = 1$ entirely, by the theorem of Duke, Friedlander and Iwaniec on Weyl sums for quadratic roots converted from smooth to sharp cut-offs; the moduli below the length in every piece, by Koksma's inequality and a weighted form of the Chinese-remainder equidistribution of Kowalski and Soundararajan, whose local input is the same for every dilation; and the pieces with $u > H^{1/2}$, where the dilations average out along u. What remains is one window in each piece with $1 < u \le H^{1/2}$, the moduli between the length and u^2 times the length, where an interval covers less than a period; it is self-dual under Hooley's reflection and it is where a power saving (Hypothesis W) is needed. Under Hypothesis W the Cesàro form of Hypothesis (E) holds for every quadratic. Over $\mathbb{F}_q[u]$ we prove that the off-diagonal is not merely bounded but has a main term: for $f = t^2 - u$ and the two smallest lengths, $\mathrm{Off}_f(N) = -2/q + O(q^{-3/2})$; a configuration count predicts, and exact computations for three discriminants and $q \le 43$ confirm, that the generic value is $-1/q$, the weight of the excluded diagonal pairs at the first degree above the cut-off, the value $-2/q$ being a small-length effect.

1 Introduction

In plain terms. Prime values of a polynomial are correlated through the small primes. Part I turned the size of that correlation into an exact identity whose main part, coming from equal roots of f modulo d, is understood, and whose remaining part, coming from pairs of distinct roots, is a question about how the square roots of the discriminant modulo composite numbers are distributed. This paper shows that the remaining part is a sum of copies of one problem, one copy for each way the two roots can agree on a part of the modulus; that the copies are of equal importance; that the first copy is a theorem of Duke, Friedlander and Iwaniec; that in every copy the moduli shorter than the length are handled by a general mixing property of the Chinese remainder theorem; that the copies with a large agreeing part average out; and that what is left is, in each copy, the moduli just above the length, where an interval covers less than one period, and where a genuinely stronger estimate is needed. Over polynomial rings the same object has an explicit main term, which we compute.

Set-up. Let $f = t^2 + bt + c \in \mathbb{Z}[t]$ be irreducible, without fixed prime divisor, with discriminant $D = b^2 - 4c$ (not a square). For a prime p let $\omega(p) = \omega_f(p)$ be the number of roots of f modulo

p: $\omega(p) = 2$ if $p \nmid 2D$ and D is a square modulo p (*split*), 0 if D is a non-square (*inert*), and ≤ 1 if $p \mid 2D$ (*special*). All moduli below are squarefree. Put

$$\lambda(p) = \frac{p}{p - 2\omega(p)}, \qquad P_p = 1 - \frac{\omega(p)^2}{(p-\omega(p))^2}, \qquad P(1) = \prod_p P_p, \qquad W_f(d) = \frac{P(1)\lambda(d)}{d}, \qquad w(u) = u\,\omega(u)\,W_f$$

λ multiplicative; this is the weight of part I, $W_f(d) = d\,b(d)\prod_{p\nmid d} P_p$ with $b(p) = (p-\omega(p))^{-2}$, since $p\,b(p)/P_p = 1/(p-2\omega(p))$. If $\omega(2) = 1$ (as for t^2+1) then $P_2 = 0$ and $\lambda(2)$ is undefined; in that case every d in the support of W_f is even, $W_f(d) = \prod_{p\mid d} p(p-\omega(p))^{-2}\prod_{p\nmid d} P_p$, and all statements below hold verbatim with the weight $w(u) = u\omega(u)W_f(u)$ (which is then supported on even u), the only property used being $W_f(ud') = W_f(u)\lambda(d')/d'$ for d' split and coprime to u. In every case $w = c_f\,g$ with $c_f > 0$ and g multiplicative on squarefree integers, $g(p) = \lambda(p)\omega(p)$ for odd p, $g(2) \in \{0, 1\}$; we write $\lambda\omega$ for g when the distinction is immaterial. For d coprime to $2D$ let $R_d = \{r \bmod d : \ r^2 \equiv D\}$ (so $|R_d| = \omega(d)$), the roots of f being $s = (r-b)/2$. Write $\langle x\rangle_d \in [1, d]$ for the representative of x mod d, and, for a length $Y > 0$ and a residue m,

$$B_d^{(Y)}(m) = \sum_{\substack{h\leq Y\\ h\equiv m\ (d)}} (Y-h) - \frac{Y^2}{2d}. \tag{1}$$

Part I, Theorem 6, proved that with $C = C(f)$

$$\sum_{h\leq H}\Big(1-\frac{h}{H}\Big)\big(S_f(h)-C^2\big) = -\tfrac12 C\log H + O_f(1) + \frac{C^2}{H}\mathrm{Off}_f^*(H), \qquad \mathrm{Off}_f^*(H) = \sum_d W_f(d) \sum_{\substack{s,s'\text{ roots of } f \bmod d\\ s\neq s'}} B_d^{(H)}(\tag{2}$$

the series converging absolutely, and that $\mathrm{Off}_f^*(H) \ll H\log H$ trivially. *Hypothesis (E) in Cesàro form* is $\mathrm{Off}_f^*(H) = o(H\log H)$; it implies Conjecture 1 of part I in Cesàro form with the leading term $-\frac12 C(f)\log H$ (part I, Theorem 6(ii) and Remark 7, since the Galois group S_2 is 2-transitive).

Results.

- *Decomposition* (Proposition 1): $\mathrm{Off}_f^*(H) = P(1)\sum_u \lambda(u)\omega(u)\,\mathcal{P}_u(H/u)$, the sum over squarefree u with $\omega(u)\geq 1$, where the piece $\mathcal{P}_u(Y)$ is the sum (2) with H replaced by Y and the roots $R_{d'}$ replaced by the roots $R_{d'}^{(u)}$ of $u^2x^2 \equiv D$, over moduli d' coprime to u all of whose primes are split. Each piece is $\ll Y$, and $\sum_u \lambda(u)\omega(u)H/u \asymp H\log H$.
- *The first piece* (Theorem 9): $\mathcal{P}_1(Y) \ll Y^{1-\delta}$ unconditionally when D is a fundamental discriminant covered by [2]; the input is a sharp-cut-off form of their Weyl-sum bound (Lemma 8).
- *Large agreeing part* (Theorem 13): $\sum_{u>H^{1/2+\varepsilon}} \lambda(u)\omega(u)\mathcal{P}_u(H/u) \ll H^{1-\delta}$, by averaging the dilations along u; the input is the distribution of $\lambda\omega$ in arithmetic progressions.
- *Small moduli of every piece* (Theorem 6): the part of $\mathcal{P}_u(Y)$ from moduli $d'\leq Y$ is $\ll Y(\log Y)^{-c}$ uniformly in u, by Koksma's inequality and a weighted version of the Chinese-remainder equidistribution theorem of Kowalski and Soundararajan [6] (Proposition 4). Hence the small-moduli part of Off_f^* is $o(H\log H)$ unconditionally.
- *The window* (§5): for $1 < u \leq H^{1/2}$ the moduli $d' \in (Y, u^2Y]$ of piece u, where the interval $[1, Y]$ is shorter than the period, carry as much mass as the small moduli, are self-dual under Hooley's reflection, and are the core of Hypothesis (E); a logarithmic saving there is not enough, and Hypothesis W is the power-saving statement that closes them. Theorem 16: under Hypothesis W, $\mathrm{Off}_f^*(H) \ll H^{1-\delta}$ for every quadratic f with D as above.

- *Function field* (Theorem 17): for $f = t^2 - u$ over $\mathbb{F}_q[u]$ and $N \in \{1,2\}$, $\mathrm{Off}_f(N) = -2/q + O(q^{-3/2})$; Conjecture 20: $\mathrm{Off}_f(N) = -1/q + O(q^{-3/2})$ whenever $\max(2N-2, \deg D) \geq N+1$, i.e. for all $N \geq 3$ and for $N \leq 2$ when $\deg D > N$, and $-2/q$ otherwise.

What is new relative to Hooley. The decomposition into pieces is implicit in Hooley's treatment of $\sum_{n \leq x} \tau(n^2 + a)$ [3], whose sum Σ_5 is our Off with the weight $1/k$ and whose Lemma 5 handles the phase that our piece $u = 1$ carries; what is new is the identification of the pieces $u > 1$ as dilated copies, their equal weight, the averaging argument for large u, the precise statement of what remains, and the function-field main term.

2 The exact decomposition

For squarefree d coprime to $2D$, the roots of f modulo d are $s = (r-b)/2$, $r \in R_d$, and $R_d = \prod_{p|d} R_p$ by the Chinese remainder theorem, each R_p being $\{\pm r_p\}$. Two roots $s \neq s'$ therefore agree exactly on a divisor $u \mid d$ and differ exactly on $d' = d/u > 1$, where every prime of d' is split; and on d' we have $r' \equiv -r$, so $s' - s \equiv -r \pmod{d'}$ and $s' - s \equiv 0 \pmod{u}$. Let $\bar{u}$ denote the inverse of u modulo d'. Then $s' - s \equiv u\bar{u} \cdot (-r) \pmod{d}$, i.e.

$$\langle s' - s \rangle_d = u \, \langle -\bar{u}r \rangle_{d'}. \tag{3}$$

If d has a special prime p ($\omega(p) = 1$) the two roots agree at p, so $p \mid u$; inert primes do not divide d at all ($\omega(d) = 0$). Define, for squarefree u with $\omega(u) \geq 1$ and squarefree d' coprime to u with all primes split,

$$R_{d'}^{(u)} = \{-\bar{u}r \bmod d' : \ r \in R_{d'}\} = \{x \bmod d' : \ u^2x^2 \equiv D \ (d')\},$$

the roots of the fixed quadratic $Q_u(x) = u^2x^2 - D$ modulo d' (the equality because $x \mapsto -\bar{u}x$ is a bijection $R_{d'} \to R_{d'}^{(u)}$ when $(u, d') = 1$), and the *piece*

$$\mathcal{P}_u(Y) = \sum_{d'} \frac{\lambda(d')}{d'} \sum_{x \in R_{d'}^{(u)}} B_{d'}^{(Y)}\big(\langle x \rangle_{d'}\big). \tag{4}$$

Proposition 1 (pieces)**.** *For every $H \geq 1$,*

$$\mathrm{Off}_f^*(H) = \sum_u w(u)\, \mathcal{P}_u(H/u), \qquad w(u) = u\, \omega(u)\, W_f(u),$$

the sum over squarefree $u \geq 1$ with $\omega(u) \geq 1$, every series converging absolutely. Moreover $\mathcal{P}_u(Y) \ll_f Y$ uniformly in u, $\sum_{u \leq t} w(u) \ll t$, and $\sum_{u \leq H} w(u)/u \asymp \log H$.

Proof. Fix $d = ud'$ as above. For $m = u\mu$ with $\mu \in [1, d']$ we have, writing $h = uh'$,

$$B_{ud'}^{(H)}(u\mu) = \sum_{\substack{h' \leq H/u \\ h' \equiv \mu \ (d')}} (H - uh') - \frac{H^2}{2ud'} = u\, B_{d'}^{(H/u)}(\mu).$$

The ordered pairs (s, s') of distinct roots modulo d with agreeing part u are in bijection with pairs (s, d'), and by (3) their difference is $u\langle -\bar{u}r \rangle_{d'}$ where r is the d'-component of $2s + b$; as s runs over the $\omega(u)\omega(d')$ roots, r runs $\omega(u)$ times over $R_{d'}$. Hence

$$\sum_{s \neq s'} B_d^{(H)}(s' - s) = \sum_{\substack{d' | d, \ d' > 1 \\ d' \ \text{split}}} \omega(u)\, u \sum_{r \in R_{d'}} B_{d'}^{(H/u)}\big(\langle -\bar{u}r \rangle_{d'}\big),$$

and with $W_f(ud') = P(1)\lambda(u)\lambda(d')/(ud')$ the claimed formula follows by summing over d; absolute convergence is inherited from (2), all terms being regrouped without change of absolute value. For the bound on $\mathcal{P}_u(Y)$: if $d' \le Y$ then, by the formula $B_{d'}^{(Y)}(m) = -Ym/d' + m^2/(2d') + Y/2 - m/2 + (d'/2)\phi(\{(Y-m)/d'\})$ of part I (proof of Theorem 6; $\phi(\theta) = \theta - \theta^2$), valid for $1 \le m \le d' \le Y$, the terms linear in m cancel between m and $d' - m$, and $B_{d'}^{(Y)}(m) + B_{d'}^{(Y)}(d'-m) = -m(d'-m)/d' + \frac{d'}{2}\big(\phi(\{\frac{Y-m}{d'}\}) + \phi(\{\frac{Y+m}{d'}\})\big) \in [-d'/4, d'/4]$; since $R_{d'}^{(u)}$ is symmetric, $|\sum_{x\in R_{d'}^{(u)}} B_{d'}^{(Y)}(\langle x\rangle_{d'})| \le \omega(d')d'/8$ (individual values $B_{d'}^{(Y)}(m)$ can be as large as $Y/2$; the symmetry of the root set is used throughout). This gives $\sum_{d'\le Y}\lambda(d')\omega(d') \ll Y$ by Shiu's theorem, since $\lambda\omega$ is multiplicative with $\lambda\omega(p) \le 2p/(p-4)$ on split primes and 0 on inert ones, of mean value $\asymp 1$. If $d' > Y$ then $B_{d'}^{(Y)}(m) = (Y-m)^+ - Y^2/(2d')$ for $m \in [1, d']$; the second term contributes $\ll Y^2\sum_{d'>Y}\lambda(d')\omega(d')/d'^2 \ll Y$, and the first contributes $\sum_{x\le Y}(Y-x)\sum_{d'>Y,\ d'|Q_u(x)}\lambda(d')/d'$; since $\sum_{d'>Y,\ d'|n} 1/d' \le n^{-1}\sum_{e|n,\ e<n/Y} e$, and on average over $x \le Y$, $\sum_{e<n/Y,\ e|Q_u(x)} e \le \sum_{e<u^2Y} e\,\rho_u(e)\,\mathbf{1}_{e|Q_u(x)}$ has mean $\ll \sum_{e<u^2Y}\rho_u(e) \ll u^2Y$ ($\rho_u(e) \le \omega(e)$ the number of roots of Q_u modulo e, of mean value $\ll 1$), while $Q_u(x) \gg u^2x^2$, the first term is $\ll \sum_{x\le Y}(Y-x)\,Y/x^2\cdot\mathbf{1}_{Q_u(x)>Y} + \sum_{x\le\sqrt{Y}/u} Y \ll Y$. Finally $\sum_{u\le t} w(u) \ll t$ and $\sum_{u\le H} w(u)/u \asymp \log H$ because $g = w/c_f$ has mean value $\asymp 1$ on squarefree integers (its Dirichlet series is $\zeta(s)L(s,\chi_D)\times$ an absolutely convergent product near $s = 1$). $\square$

In words. The off-diagonal is a sum over u of the same object, evaluated at length H/u, for the roots of $u^2x^2 \equiv D$ instead of $x^2 \equiv D$; since $\sum_u w(u)/u$ grows like $\log H$ while each piece is $O(H/u)$, the pieces are equally important on a logarithmic scale, and no range of u can be discarded. The dilation by $\bar{u}$ is the whole difficulty: for $u = 1$ the roots are those of a fixed fundamental quadratic; for $u > 1$ they are those of a quadratic whose discriminant $4u^2D$ grows with u, and existing theorems degrade with the discriminant.

3 Reduction of a piece to Weyl sums

Fix u and write Y for the length, $R'_{d'} = R_{d'}^{(u)}$, $\rho(d') = |R'_{d'}|$. Split $\mathcal{P}_u(Y) = \mathcal{P}_u^{\le}(Y) + \mathcal{P}_u^{>}(Y)$ according to $d' \le Y$ or $d' > Y$.

3.1 Small moduli: Fourier expansion and partial summation

For $d' \le Y$ and $m \in [1, d']$, with $\phi(\theta) = \theta - \theta^2$ (part I, proof of Theorem 6),

$$B_{d'}^{(Y)}(m) = -\frac{Ym}{d'} + \frac{m^2}{2d'} + \frac{Y}{2} - \frac{m}{2} + \frac{d'}{2}\phi\Big(\Big\{\frac{Y-m}{d'}\Big\}\Big),$$

and summing over the symmetric set $R'_{d'}$ ($x \in R' \Rightarrow -x \in R'$, so m and $d' - m$ occur together) gives $\sum_{x\in R'} B_{d'}^{(Y)}(\langle x\rangle_{d'}) = \sum_{\{x,-x\}}\big[-\frac{m(d'-m)}{d'} + \frac{d'}{2}\big(\phi(\{\frac{Y-m}{d'}\}) + \phi(\{\frac{Y+m}{d'}\})\big)\big]$. Since $\phi(\{\theta\}) = \frac{1}{6} - \pi^{-2}\sum_{k\ge1}k^{-2}\cos 2\pi k\theta$ and $\frac{1}{6} - \theta(1-\theta) = \pi^{-2}\sum_{k\ge1}k^{-2}\cos 2\pi k\theta$,

$$\sum_{x\in R'_{d'}} B_{d'}^{(Y)}(\langle x\rangle_{d'}) = \frac{d'}{2\pi^2}\sum_{k\ge1}\frac{1}{k^2}\big(1 - \cos\tfrac{2\pi kY}{d'}\big)\,\rho_k(d'), \qquad \rho_k(d') = \sum_{x\in R'_{d'}} \mathrm{e}\Big(\frac{kx}{d'}\Big), \tag{5}$$

an absolutely convergent expansion for each d' ($\rho_k(d')$ is real by symmetry). Hence

$$\mathcal{P}_u^{\le}(Y) = \frac{1}{2\pi^2}\sum_{k\ge1}\frac{1}{k^2}T_k(Y), \qquad T_k(Y) = \sum_{d'\le Y}\lambda(d')\rho_k(d')\big(1 - \cos\tfrac{2\pi kY}{d'}\big), \tag{6}$$

where now the interchange is legitimate because d' ranges over a finite set, and trivially $|T_k(Y)| \le 2\sum_{d'\le Y}\lambda\rho \ll Y$.

Lemma 2 (from sharp Weyl sums to twisted sums)**.** *Suppose that for some $\theta < 1$, $B \geq 0$ and all $k \geq 1$, $t \geq 2$,*

$$S_k(t) := \sum_{d' \leq t} \lambda(d')\rho_k(d') \ll (uk)^B\, t^\theta. \tag{7}$$

Then for every $Y_0 \in [1, Y]$,

$$T_k(Y) \ll Y_0 \log Y_0 + (uk)^B \big(Y^\theta + kY\, Y_0^{\theta-1}\big) \log Y.$$

In particular, with $Y_0 = (kY)^{1/(2-\theta)}$, $T_k(Y) \ll (uk)^B (kY)^{1/(2-\theta)} \log Y$.

Proof. The moduli $d' \leq Y_0$ contribute at most $2\sum_{d' \leq Y_0} \lambda\rho \ll Y_0 \log Y_0$ (crudely). On a dyadic block $V < d' \leq 2V$, $V \geq Y_0$, partial summation against S_k with the function $g(t) = 1 - \cos(2\pi kY/t)$, which satisfies $|g| \leq 2$ and $|g'(t)| \leq 2\pi kY/t^2$, gives $\big|\sum_{V<d'\leq 2V} \lambda\rho_k\, g\big| \leq |S_k(2V)g(2V)| + |S_k(V)g(V)| + \int_V^{2V} |S_k(t)g'(t)|\, dt \ll (uk)^B V^\theta (1 + kY/V)$. Summing over the dyadic $V \in [Y_0, Y]$ gives the claim. □

In words. The phase $\cos(2\pi kY/d')$ oscillates in d' once $d' \ll kY$, and the cost of integrating it against a partial sum with a power saving is a factor kY/d'; the small moduli, where that factor is large, are handled trivially, and the balance point $Y_0 = (kY)^{1/(2-\theta)}$ leaves a power saving. Nothing more than a sharp-cut-off Weyl bound with polynomial dependence on the frequency is used.

3.2 Small moduli without the phase: Koksma and the mixing of the Chinese remainder theorem

The Fourier route makes the phase $\cos(2\pi kY/d')$ appear; the discrepancy route does not. For $d' \leq Y$ and the symmetric set $R'_{d'}$, the identity before (5) reads

$$\sum_{x \in R'_{d'}} B_{d'}^{(Y)}(\langle x\rangle_{d'}) = \frac{d'}{2} \sum_{x \in R'_{d'}} \Psi_{Y/d'}\Big(\Big\{\frac{x}{d'}\Big\}\Big), \qquad \Psi_\eta(\theta) = -\theta(1-\theta) + \tfrac{1}{2}\big(\phi(\{\eta - \theta\}) + \phi(\{\eta + \theta\})\big),$$

and Ψ_η has mean 0 on $[0, 1)$ and total variation ≤ 3, uniformly in η. Koksma's inequality therefore gives

$$\Big|\sum_{x \in R'_{d'}} B_{d'}^{(Y)}(\langle x\rangle_{d'})\Big| \leq \tfrac{3}{2}\, d'\, \omega(d')\, \mathrm{disc}(\Delta_{d'}^{(u)}), \tag{8}$$

where $\Delta_{d'}^{(u)}$ is the probability measure $\omega(d')^{-1} \sum_{x \in R'_{d'}} \delta_{\{x/d'\}}$ and disc its discrepancy, $\sup_I |\Delta(I) - |I||$ over intervals $I \subset \mathbb{R}/\mathbb{Z}$. The point is that (8) does not see the length Y at all.

We shall need a weighted form of Rankin's bound for smooth numbers, with Dickman-type decay.

Lemma 3 (weighted smooth numbers)**.** *Let $w \geq 0$ be multiplicative, supported on squarefree integers, with $w(p) \leq C_0$ for all primes p, where $C_0 \geq 1$. There are an absolute constant z_0 and $u_0 = 16eC_0^2$ such that for $z \geq z_0$, $X \geq 1$ and $u := \log X / \log z \in [u_0, \log z]$,*

$$\sum_{\substack{s > X \\ P^+(s) \leq z}} \frac{w(s)}{s} \;\leq\; \exp\Big(-\frac{u \log u}{2}\Big) \prod_{p \leq z} \Big(1 + \frac{w(p)}{p}\Big).$$

Proof. For $\delta > 0$ (Rankin), the sum is at most $X^{-\delta} \sum_{P^+(s) \leq z} w(s) s^{\delta-1} = X^{-\delta} \prod_{p \leq z} (1 + w(p)p^{\delta-1})$, and $1 + ap^\delta \leq (1 + a)\exp(a(p^\delta - 1))$ for $a \geq 0$ gives

$$\prod_{p \leq z} \Big(1 + \frac{w(p)}{p^{1-\delta}}\Big) \leq \prod_{p \leq z} \Big(1 + \frac{w(p)}{p}\Big) \exp(C_0 E), \qquad E = \sum_{p \leq z} \frac{p^\delta - 1}{p}.$$

Put $\lambda = \delta \log z$ and $g(v) = (e^{\delta v} - 1)/v$, an increasing function of $v > 0$, so that $(p^\delta - 1)/p = g(\log p) \log p/p$. With $S(t) = \sum_{p \le t} \log p/p = \log t + R(t)$, $|R(t)| \le R_0$ for $t \ge 2$ (Mertens), Stieltjes integration gives

$$E = \int_{2^-}^{z} g(\log t)\, dS(t) = \int_{\log 2}^{\log z} g(v)\, dv + \int_{2^-}^{z} g(\log t)\, dR(t) \le \int_0^\lambda \frac{e^s - 1}{s}\, ds + 3R_0\, g(\log z) + g(\log 2) \log 2,$$

where we used that g is increasing to bound $\int R\, dg$. Since $(e^s - 1)/s \le e^s$, the integral is $\le e^\lambda$; the second term is $\le 3R_0 e^\lambda/\log z$; and $g(\log 2)\log 2 = 2^\delta - 1 \le 1$ as $\delta \le 1$ (because $\lambda \le \log u \le \log\log z$). Hence $E \le 2e^\lambda$ once $\log z \ge 6R_0$ and $e^\lambda \ge 2$. Now choose $\lambda = \log u - \log(4C_0)$, which is $\ge \log 4$ for $u \ge 16C_0$ and is $\le \log u \le \log z$ as required: then $e^\lambda = u/(4C_0)$, so $C_0 E \le u/2$, while $\delta \log X = \lambda u = u \log u - u \log(4C_0)$. The exponent is therefore at most $-u \log u + u \log(4C_0) + u/2 \le -\frac{1}{2} u \log u$ as soon as $\frac{1}{2}\log u \ge \log(4C_0) + \frac{1}{2}$, i.e. $u \ge 16eC_0^2$. $\square$

Proposition 4 (weighted Kowalski–Soundararajan)**.** *Let $(A_p)_p$ be a family of subsets $A_p \subset \mathbb{Z}/p$, $\rho(p) = |A_p|$, let Q be the set of squarefree q all of whose prime factors have $\rho(p) \ge 1$, and for $q \in Q$ let $A_q \subset \mathbb{Z}/q$ be the set of classes reducing into A_p for every $p \mid q$, $\Delta_q = \rho(q)^{-1} \sum_{a \in A_q} \delta_{\{a/q\}}$, $\rho(q) = \prod_{p \mid q} \rho(p)$. Let $w \ge 0$ be a multiplicative function supported on squarefree integers with $w(p) \le C_0$ ($C_0 \ge 1$) and $w(p) = 0$ unless $\rho(p) \ge 1$, and suppose that for some $\beta, \gamma > 0$ and $t_0 \ge 3$*

$$\sum_{\substack{p \le t \\ \rho(p) \ge 2}} \frac{w(p)}{p} \;\ge\; \beta \log\log t \quad (t \ge t_0), \qquad\qquad \sum_{p \le t} \frac{w(p)}{p} \;\le\; \log\log t + \gamma \quad (t \ge 3). \tag{9}$$

Then there are $c = c(\beta) > 0$ and $x_1 = x_1(\beta, \gamma, C_0, t_0)$ such that for $x \ge x_1$

$$\sum_{\substack{q \le x \\ q \in Q}} w(q)\, \mathrm{disc}(\Delta_q) \;\le\; x\,(\log x)^{-c}.$$

One can take $c = \beta(1 - 2^{-1/2})/4$. The constants do not depend on the sets A_p, only on their sizes through (9)*. (By the second condition, $S_w(x) = \sum_{q \le x,\, q \in Q} w(q) \ll_{\gamma, C_0} x$, so the bound is a saving of* $(\log x)^{-c}$ *over the trivial one whenever w has positive mean value.)*

Proof. We follow [6, §§3–4] and indicate the changes. Write $W(h; q) = \rho(q)^{-1} \sum_{a \in A_q} \mathrm{e}(ha/q)$ for the normalised Weyl sums. Two facts from [6, Lemma 3.5, Lemma 3.7] hold for every family (A_p), with no hypothesis on the sets beyond their sizes: (a) for coprime q_1, q_2, $W(h; q_1 q_2) = W(\bar{q}_1 h; q_2) W(\bar{q}_2 h; q_1)$ with $q_1 \bar{q}_1 \equiv 1\ (q_2)$, $q_2 \bar{q}_2 \equiv 1\ (q_1)$; (b) for $q \in Q$ and $L \ge 2$,

$$\frac{1}{q} \sum_{a \bmod q} \sum_{0 < |h| \le L} \frac{|W(ah; q)|}{|h|} \;\ll\; (\log L) \prod_{p \mid q} \Big(\frac{1}{\sqrt{\rho(p)}} + \frac{1}{\sqrt{p}} \Big), \tag{10}$$

which is their Lemma 3.7 in dimension one, where the hyperplane constant $\lambda(p)$ equals 1 (a point). Also, by the Erdős–Turán inequality, $\mathrm{disc}(\Delta_q) \ll L^{-1} + \sum_{0 < |h| \le L} |W(h; q)|/|h|$.

Throughout, $\prod_{p \le t}(1 + w(p)/p) \le \exp(\sum_{p \le t} w(p)/p) \le e^\gamma \log t$ by (9), and $\sum_{t_1 < p \le t_2} w(p)/p \le C_0 \sum_{t_1 < p \le t_2} 1/p \le C_0 \log(\log t_2/\log t_1) + O(C_0)$ by Mertens.

Let $P = \sum_{p \le x,\ \rho(p) \ge 2} w(p)(1 - \rho(p)^{-1/2})/p$, so that $\beta(1 - 2^{-1/2}) \log\log x \le P \le \log\log x + \gamma$ by (9), and put $z = x^{1/P}$, so that $\log z = \log x/P \ge \log x/(2 \log\log x)$ for $x \ge x_1$. Factor $q = rs$ with s the z-smooth and r the z-rough part; $w(q) = w(r) w(s)$.

Case $s \le x^{1/3}$. Fix s and L. By (a) and the Erdős–Turán inequality, for r coprime to s, $\mathrm{disc}(\Delta_{rs}) \ll L^{-1} + \sum_{0 < |h| \le L} |W(\bar{r} h; s)|/|h|$, where we bounded $|W(\bar{s} h; r)| \le 1$. Group the r by the class $a \equiv \bar{r} \pmod{s}$: then

$$\sum_{\substack{r \le x/s,\ rs \in Q \\ p \mid r \Rightarrow p > z}} w(r)\, \mathrm{disc}(\Delta_{rs}) \;\ll\; \sum_{\substack{a \bmod s \\ (a,s) = 1}} \Big(\frac{1}{L} + \sum_{0 < |h| \le L} \frac{|W(ah; s)|}{|h|} \Big) \sum_{\substack{r \le x/s,\ r \equiv \bar{a}\ (s) \\ p \mid r \Rightarrow p > z}} w(r).$$

This is where the first change occurs. The inner sum is a sum of the non-negative multiplicative function $w \cdot \mathbf{1}_{z\text{-rough}}$ (which is $\le C_0^{\omega(n)} \ll_\varepsilon n^\varepsilon$) over a progression to the modulus $s \le x^{1/3} \le (x/s)^{1/2}$; by Shiu's theorem [7], uniformly in a,

$$\sum_{\substack{r\le x/s,\ r\equiv\bar a\ (s)\\ p|r\Rightarrow p>z}} w(r) \;\ll\; \frac{x/s}{\varphi(s)\log(x/s)}\exp\Big(\sum_{z<p\le x}\frac{w(p)}{p}\Big) \;\ll\; \frac{x}{s\,\varphi(s)\log x}\,P^{C_0},$$

since $\sum_{z<p\le x} w(p)/p \le C_0 \log P + O(C_0)$. Inserting (10),

$$\sum_{\substack{r\le x/s\\ rs\in Q,\ r\ \mathrm{rough}}} w(r)\,\mathrm{disc}(\Delta_{rs}) \;\ll\; \frac{xP^{C_0}}{\varphi(s)\log x}\Big(\frac{1}{L} + (\log L)\prod_{p|s}\Big(\frac{1}{\sqrt{\rho(p)}}+\frac{1}{\sqrt{p}}\Big)\Big).$$

Now multiply by $w(s)$ and sum over z-smooth $s \le x^{1/3}$, $s \in Q$. Since $w(s)/\varphi(s) = \prod_{p|s} w(p)/(p-1)$,

$$\sum_s \frac{w(s)}{\varphi(s)}\prod_{p|s}\Big(\frac{1}{\sqrt{\rho(p)}}+\frac{1}{\sqrt{p}}\Big) \le \prod_{\substack{p\le z\\ p\in Q}}\Big(1+\frac{w(p)}{p-1}\Big(\frac{1}{\sqrt{\rho(p)}}+\frac{1}{\sqrt{p}}\Big)\Big) \ll \prod_{\substack{p\le z\\ p\in Q}}\Big(1+\frac{w(p)}{p}\Big)\exp\Big(-\sum_{\substack{p\le z\\ \rho(p)\ge 2}}\frac{w(p)}{p}\Big(1-\frac{1}{\sqrt{\rho(p)}}\Big)\Big)$$

because $(1+w(1/\sqrt{\rho}+p^{-1/2})/(p-1))/(1+w/p) \le \exp(-w(1-1/\sqrt{\rho})/p + O(C_0 p^{-3/2}))$ and the primes with $\rho(p)=1$ contribute $O(1)$; the exponential is $\exp(-P_z)$ with P_z the sum defining P cut at z, and $P - P_z = \sum_{z<p\le x}(\dots) \le C_0 \log P + O(C_0)$, so $\exp(-P_z) \ll P^{C_0} e^{-P}$. Similarly $\sum_s w(s)/\varphi(s) \ll \prod_{p\le z,\ p\in Q}(1+w(p)/p)$. With $L = e^P$ the case contributes

$$\ll \frac{x}{\log x}\prod_{\substack{p\le z\\ p\in Q}}\Big(1+\frac{w(p)}{p}\Big)\cdot P^{2C_0+1}e^{-P} \;\ll\; x\,P^{2C_0+1}e^{-P},$$

using $\prod_{p\le z}(1+w(p)/p) \le e^\gamma \log z \le e^\gamma \log x$.

Case $s > x^{1/3}$. Here we use disc ≤ 1 and bound the weighted number of $q \le x$ in Q whose z-smooth part exceeds $x^{1/3}$. For $X \ge 1$ let $\mathcal{R}(X) = \sum_{r\le X,\ p|r\Rightarrow p>z} w(r)$. If $X < z$ then $\mathcal{R}(X) = 1$. If $X \ge z$, the upper bound of Halberstam and Richert for non-negative multiplicative functions [8, Theorem III.3.5] gives $\mathcal{R}(X) \ll_{C_0} (X/\log X)\prod_{z<p\le X}(1+w(p)/p) \ll (X/\log z)P^{C_0} = XP^{C_0+1}/\log x$. In both cases $\mathcal{R}(X) \le 1 + O(XP^{C_0+1}/\log x)$, and for $X = x/s \ge 1$ this is $\le 2x/s$ once $x \ge x_1$. Hence

$$\sum_{\substack{q\le x,\ q\in Q\\ s>x^{1/3}}} w(q) \;\le\; \sum_{\substack{x^{1/3}<s\le x\\ P^+(s)\le z}} w(s)\,\mathcal{R}(x/s) \;\le\; 2x\sum_{\substack{s>x^{1/3}\\ P^+(s)\le z}}\frac{w(s)}{s}.$$

Apply Lemma 3 with $X = x^{1/3}$, so $u = \log X/\log z = P/3$; the conditions $z \ge z_0$ and $u_0 \le u \le \log z$ hold for $x \ge x_1(\beta,\gamma,C_0,t_0)$, because $P \to \infty$ and $P/3 \le \log\log x \le \log z$. The sum is therefore $\le e^\gamma \log z \cdot \exp(-\frac{1}{6}P\log(P/3))$, and since $\log z \le \log x = e^{\log\log x} \le e^{P/(\beta(1-2^{-1/2}))}$,

$$\sum_{\substack{q\le x,\ q\in Q\\ s>x^{1/3}}} w(q) \;\ll_\gamma\; x\exp\Big(\frac{P}{\beta(1-2^{-1/2})} - \frac{P\log(P/3)}{6}\Big) \;\le\; x\,e^{-P}$$

as soon as $\log(P/3) \ge 6(1+1/(\beta(1-2^{-1/2})))$, which again holds for $x \ge x_1$. This is the second change: the Dickman-type decay of Lemma 3, with $u \asymp \log\log x$, beats every fixed power of $\log x$, so no analogue of the counting argument of [6, Lemma 3.2] is needed.

Conclusion. Adding the two cases, $\sum_{q\le x,\ q\in Q} w(q)\mathrm{disc}(\Delta_q) \ll_{\gamma,C_0} x\,P^{2C_0+1}e^{-P}$, and since $P \ge \beta(1-2^{-1/2})\log\log x$, this is $\le x(\log x)^{-c}$ with $c = \beta(1-2^{-1/2})/4$ for $x \ge x_1$. □

In words. Kowalski and Soundararajan's theorem says that a set defined prime by prime through the Chinese remainder theorem is equidistributed modulo most q, at a rate governed only by how many primes carry at least two residues; the proof averages the rough part of the modulus over residue classes modulo the smooth part, and the local sets enter only through their sizes. Multiplicative weights of bounded mean ride along: Shiu's theorem replaces the sieve count, Rankin's trick with Dickman decay disposes of the moduli with a large smooth part, and the rest is bookkeeping in Euler products.

Remark 5. For $w = \lambda\omega$ and the split primes of a quadratic f: $w(p) = 2p/(p-4) \in [2, 10]$ for split $p \geq 5$, $w(p) = 0$ at inert primes and at the finitely many primes dividing $2D$, $\rho(p) = 2$ at split primes, and (9) holds with $\beta = 1 - o(1)$ and some $\gamma = \gamma(f)$ by the prime number theorem for the character χ_D ($\sum_{p \leq t \text{ split}} 1/p = \frac{1}{2}\log\log t + O_D(1)$). Removing in addition the primes dividing a squarefree u (where $R_p^{(u)} = \emptyset$) lowers the first sum in (9) by at most $C_0 \sum_{p|u} 1/p \leq C_0(\log\log\log u + O(1))$ and can only lower the second; so (9) holds with $\beta = \frac{1}{2}$, the same γ, and $t_0 = \exp((\log\log u)^A)$ for an absolute $A = A(C_0)$: in particular uniformly for $t \geq u \geq u_1$. The resulting exponent is $c = (1 - 2^{-1/2})/8 \approx 0.037$; no attempt has been made to optimise it, and none is needed for the Cesàro form.

Theorem 6 (small moduli, unconditional)**.** *For every irreducible quadratic f there is $c = c(f) > 0$ such that, uniformly for squarefree u with $\omega(u) \geq 1$ and $Y \geq \max(u, Y_1(f))$,*

$$\mathcal{P}_u^{\leq}(Y) = \sum_{d' \leq Y} \frac{\lambda(d')}{d'} \sum_{x \in R_{d'}^{(u)}} B_{d'}^{(Y)}(\langle x \rangle_{d'}) \ll_f Y(\log Y)^{-c}.$$

Consequently $\sum_{u \leq H^{1/2}} \lambda(u)\omega(u)\, \mathcal{P}_u^{\leq}(H/u) \ll_f H(\log H)^{1-c}$: the small-moduli part of the pieces with $u \leq H^{1/2}$ is $o(H \log H)$ unconditionally (the pieces with $u > H^{1/2}$ are treated in Theorem 13).

Proof. By (8), $|\mathcal{P}_u^{\leq}(Y)| \leq \frac{3}{2} \sum_{d' \leq Y} \lambda(d')\omega(d') \operatorname{disc}(\Delta_{d'}^{(u)})$, the sum over squarefree split d' coprime to u. Apply Proposition 4 to the family $A_p = R_p^{(u)}$ (of size 2 at split $p \nmid 2Du$, empty otherwise) and to $w = \lambda\omega \cdot \mathbf{1}_{(\cdot, 2Du)=1}$; by Remark 5 the hypotheses hold with constants depending only on f provided $Y \geq u \geq u_1$, and for the finitely many $u < u_1$ they hold with $t_0 = t_0(f)$. Since Q is then exactly the set of admissible d' and $\rho(d') = \omega(d')$, the bound is $\ll Y(\log Y)^{-c}$ with $c = c(f)$. For the consequence, the weight $\lambda\omega(u)$ has $\sum_{u \leq U} \lambda\omega(u)/u \ll \log U$, and for $u \leq H^{1/2}$ we have $Y = H/u \geq H^{1/2} \geq u$, so $\sum_{u \leq H^{1/2}} \lambda\omega(u)\,(H/u)(\log(H/u))^{-c} \ll H(\log H)^{-c} \sum_{u \leq H^{1/2}} \lambda\omega(u)/u \ll H(\log H)^{1-c}$. ◻

In words. For moduli shorter than the length there is no short-interval problem, only the question whether the dilated roots are spread out modulo d'; and spreading out is a consequence of the Chinese remainder theorem alone, as Kowalski and Soundararajan showed, at a rate that does not see which two-point sets sit at the primes. So this half of every piece is unconditional, and the loss against the trivial bound is a power of log, which is exactly what the Cesàro form can afford.

3.3 Large moduli: the far moduli are trivial

For $d' > Y$ and $m \in [1, d']$ at most one $h \leq Y$ lies in the class of m, so $B_{d'}^{(Y)}(m) = (Y - m)^+ - Y^2/(2d')$ and

$$\sum_{d' > Z} \frac{\lambda(d')}{d'} \sum_{x \in R'_{d'}} B_{d'}^{(Y)}(\langle x \rangle_{d'}) = \sum_{x \leq Y} (Y - x) \sum_{\substack{d' > Z \\ d' | Q_u(x)}} \frac{\lambda(d')}{d'} - \frac{Y^2}{2} \sum_{d' > Z} \frac{\lambda(d')\rho(d')}{d'^2} =: \mathcal{A}_Z - \mathcal{E}_Z \qquad (Z \geq Y), \tag{11}$$

where d' ranges over squarefree split moduli coprime to u, and we used that an integer $x \in [1, Y]$ is the representative of an element of $R'_{d'}$ exactly when $d' \mid Q_u(x)$. Both terms are of the same size, and the question is how well they cancel. The next lemma says that for $Z = u^2 Y$ there is nothing to cancel: both terms are already small.

Lemma 7 (far moduli)**.** *Let u be squarefree with $\omega(u) \geq 1$ and $Y \geq 4|D| + 4$. Then*

$$\mathcal{P}_u^{\mathrm{far}}(Y) := \sum_{d' > u^2 Y} \frac{\lambda(d')}{d'} \sum_{x \in R_{d'}^{(u)}} B_{d'}^{(Y)}(\langle x \rangle_{d'}) \ll_f \frac{Y}{u^2},$$

uniformly in u. Consequently $\sum_{u > u_1} w(u) \, |\mathcal{P}_u^{\mathrm{far}}(H/u)| \ll_f H/u_1$ for every $u_1 \geq 1$.

Proof. Put $Z = u^2 Y$. Since $\lambda\rho$ has mean value $\ll 1$ (Proposition 1), $\sum_{d' > Z} \lambda\rho(d')/d'^2 \ll 1/Z$ and $\mathcal{E}_Z \ll Y^2/Z = Y/u^2$.

For $\mathcal{A}_Z$ write $\lambda = 1 * \kappa$ with κ multiplicative, supported on squarefree split integers, $\kappa(p) = \lambda(p) - 1 = 4/(p-4) \geq 0$. Dropping the conditions on the moduli (all terms are non-negative) and substituting $d' = ge$... more precisely writing $d' = g\, d''$ with $g \mid d'$ carrying κ, and $d'' = n/(ge)$ with $n = Q_u(x)$, so that $1/d'' = ge/n$ and $d'' > Z/g$ becomes $e < n/(gZ)$,

$$\sum_{\substack{d' > Z \\ d' \mid n}} \frac{\lambda(d')}{d'} \leq \sum_{g \mid n} \kappa(g) \sum_{\substack{e \mid n/g \\ e < n/(gZ)}} \frac{e}{n}, \qquad \text{hence} \qquad \mathcal{A}_Z \leq \sum_g \kappa(g) \sum_e e \sum_{\substack{x \leq Y,\ ge \mid Q_u(x) \\ Q_u(x) > geZ}} \frac{Y - x}{Q_u(x)}.$$

In the inner sum $Q_u(x) > geu^2 Y$ and $Q_u(x) \leq u^2 Y^2 + |D| \leq 2u^2 Y^2$, so $ge < 2Y$; moreover $u^2 x^2 = Q_u(x) + D > geu^2 Y - |D| \geq geu^2 Y/2$ because $Y \geq 2|D|$, so $x > x_0 := \sqrt{geY/2}$, and then $u^2 x^2 \geq 2|D|$ gives $Q_u(x) \geq u^2 x^2/2$. Therefore

$$\frac{Y - x}{Q_u(x)} \leq \min\Big(\frac{1}{geu^2}, \ \frac{2Y}{u^2 x^2} \Big).$$

The x with $ge \mid Q_u(x)$ lie in $\rho_u(ge)$ residue classes modulo ge, where $\rho_u(m)$ denotes the number of roots of Q_u modulo m (for all m, not only squarefree split ones). In one class, the first term beyond x_0 contributes at most $2Y/(u^2 x_0^2) = 4/(geu^2)$ and the others at most $(ge)^{-1} \int_{x_0}^\infty 2Y \, dx/(u^2 x^2) = 2\sqrt{2}\, \sqrt{Y/(ge)}/(geu^2)$; so

$$\mathcal{A}_Z \leq \frac{1}{u^2} \sum_g \frac{\kappa(g)}{g} \sum_{e < 2Y/g} \rho_u(ge) \Big(4 + 3\sqrt{\frac{Y}{ge}} \Big).$$

Now ρ_u is multiplicative; $\rho_u(p^k) = 1 + \chi_D(p) \leq 2$ for $p \nmid 2Du$; $\rho_u(p^k) = 0$ for $p \mid u$, $p \nmid D$; and for the finitely many $p \mid 2D$, $\rho_u(p^k) \leq 4p^{v_p(D)+2}$ uniformly in k and in squarefree u (if $k > v_p(D) + 2$ a solution forces $v_p(u^2 x^2) = v_p(D)$, which fixes $v_p(x)$ and leaves a congruence $x'^2 \equiv D'$ $(p^{k - v_p(D)})$ with D' a unit, having at most 4 solutions, each lifting to $p^{v_p(D)/2+1}$ classes). Hence $\rho_u(ge) \leq \rho_u(g)\rho_u(e)$ for squarefree split g, and $\sum_{e \leq t} \rho_u(e) \ll_f t$ uniformly in u (the Dirichlet series of ρ_u is $\zeta(s) L(s, \chi_D)$ times a product absolutely convergent for $\Re s > \frac{1}{2}$ with constants depending only on f), whence $\sum_{e \leq t} \rho_u(e) e^{-1/2} \ll_f t^{1/2}$ by partial summation. The inner sum is therefore $\ll \rho_u(g) Y/g$, and $\mathcal{A}_Z \ll (Y/u^2) \sum_g \kappa(g)\rho_u(g)/g^2 \ll Y/u^2$.

For the consequence, $\sum_{u \leq t} w(u) \ll t$ (Proposition 1), so $\sum_{u > u_1} w(u) u^{-3} \ll u_1^{-2}$... more precisely $\sum_{u > u_1} w(u)(H/u)u^{-2} = H \sum_{u > u_1} w(u) u^{-3} \ll H/u_1^2 \leq H/u_1$. □

In words. When the modulus exceeds u^2 times the length, each root that falls below the length has a complementary divisor $e = Q_u(x)/d'$ below the length, and the weights $1/d' = e/Q_u(x)$ decay like $1/x^2$; counting crudely, the whole far range weighs only Y/u^2, which is summable over u. So the far moduli of the pieces with large agreeing part are negligible for trivial reasons, and only for bounded u, where Y/u^2 is not small compared with Y, is an actual cancellation needed; that case is Hooley's reflection and is treated in Proposition **??**. The trivial bound also fixes the scale of everything else: the small moduli $d' \le Y$ and the window $Y < d' \le u^2 Y$ each have trivial size $\asymp Y$, the far moduli Y/u^2.

4 The first piece and the large pieces

4.1 $u = 1$: the theorem of Duke, Friedlander and Iwaniec

Lemma 8 (sharp cut-offs from smooth ones)**.** *Let D be a fundamental discriminant for which Theorem 1.1 of [2] holds (positive odd D as stated there; negative D by their §16), and let $q \ge 1$, $k \ge 1$. Then for $t \ge 2$,*

$$\sum_{\substack{d \le t \\ q \mid d}} \sum_{r \in R_d} \mathrm{e}\Big(\frac{kr}{d}\Big) \ll_{D,\varepsilon} k^{1/4}\, t^{11/12+\varepsilon},$$

uniformly in q; the same holds with the additional factor $\lambda(d)$, and with d restricted to squarefree split moduli.

Proof. Theorem 1.1 of [2] states: for f smooth, supported on $[V, 2V]$, with $|f| \le 1$ and $y^2|f''(y)| \le 1$, $\sum_{c \equiv 0(q)} f(c) \sum_{b^2 \equiv D(c)} \mathrm{e}(hb/c) \ll h^{1/4}(V + h\sqrt{D})^{3/4} D^{1/8-1/1331}$ with an absolute implied constant. The statement is linear in f; hence if f satisfies $|f| \le \Delta^2$, $y^2|f''| \le \Delta^2$, the bound holds with an extra factor Δ^2 (apply it to f/Δ^2). Let $\Delta = t^{1/12}$ and let χ_t be a smooth function equal to 1 on $[0, t(1-1/\Delta)]$, to 0 on $[t, \infty)$, with $|\chi_t^{(j)}| \ll (\Delta/t)^j$. Decompose $\chi_t = \sum_V \chi_t \psi_V$ with a smooth dyadic partition of unity $\{\psi_V\}$ on $[1, \infty)$: each piece is supported on $[V, 2V]$ with $y^2|(\chi_t\psi_V)''| \ll 1$ except the piece containing the edge, for which $y^2|\cdot''| \ll \Delta^2$. Summing the DFI bound over the dyadic $V \le t$ gives $\ll \Delta^2 k^{1/4} t^{3/4+\varepsilon}$ for fixed D. The difference between the sharp cut-off and χ_t is supported on $(t(1-1/\Delta), t]$ and is bounded, for the moduli $q \mid d$ there, by $\sum_{d \in (t(1-1/\Delta), t],\ q \mid d} \omega(d) \ll (t/(q\Delta) + 1)\, t^{\varepsilon}$ by Shiu's theorem for the multiplicative function ω on short intervals of progressions ($t/\Delta \ge t^{\varepsilon}$). Altogether $\ll t^{1/6} k^{1/4} t^{3/4+\varepsilon} + t^{11/12+\varepsilon} \ll k^{1/4} t^{11/12+\varepsilon}$. The weight $\lambda(d) = \sum_{g \mid d} \kappa(g)$ is removed by writing $\sum_d \lambda(d) F(d) = \sum_g \kappa(g) \sum_{g \mid d} F(d)$, applying the bound with q replaced by $\mathrm{lcm}(q, g)$, and using $\sum_g \kappa(g) < \infty$; the restriction to squarefree split d is a further Möbius inversion (over $\ell^2 \mid d$ and over the inert and special primes, which carry no roots or a single root and are removed by inclusion–exclusion at the cost of the divisor bound). $\square$

Theorem 9 (the first piece: small moduli)**.** *For D as in Lemma 8, $\mathcal{P}_1^{\le}(Y) = \sum_{d' \le Y} \frac{\lambda(d')}{d'} \sum_{x \in R_{d'}} B_{d'}^{(Y)}(\langle x \rangle_{d'}) \ll$ $Y^{12/13+\varepsilon}$. Together with Proposition* **??** *(the far moduli, $u = 1$) this gives $\mathcal{P}_1(Y) = o(Y)$ unconditionally; the window is empty for $u = 1$.*

Proof. Lemma 8 gives (7) for $u = 1$ with $\theta = \frac{11}{12} + \varepsilon$, $B = \frac{1}{4}$. By Lemma 2, $T_k(Y) \ll k^{1/4}(kY)^{12/13+\varepsilon}$, and inserting into (6): $\sum_{k \le K} k^{-2} T_k \ll Y^{12/13+\varepsilon} \sum_{k \le K} k^{-2+1/4+12/13} \ll Y^{12/13+\varepsilon} K^{0.18}$, while $\sum_{k > K} k^{-2}|T_k| \ll Y/K$; with $K = Y^{1/20}$ both are $\ll Y^{12/13+\varepsilon}$. $\square$

Remark 10. A power saving for the far moduli of the first piece is within reach of the same input: after the reflection of Proposition **??** the moduli are again below the length, and Lemma 8 applies to the sharp sums $\sum_{e \le t} \rho_j(ge)$ (the divisor g playing the role of q), with the two phases of the reflected sum handled by partial summation as in Lemma 2. We have not carried this out, because the first piece is a single term of size $O(H)$ in a sum whose target is $o(H \log H)$, and because the identification of the reflected main term (Proposition **??**, Step 4) is the same in both versions.

4.2 Large u: averaging the dilations

Throughout this subsection $g = w/c_f$ is the multiplicative function of the set-up ($g(p) = \lambda(p)\omega(p)$ for odd $p \nmid D$, g supported on squarefree integers coprime to D), $\chi = \chi_D$ is the Kronecker symbol, and m, m_d are the constants

$$m = L(1,\chi)\prod_p\Big(1-\frac{1}{p}\Big)\Big(1-\frac{\chi(p)}{p}\Big)\Big(1+\frac{g(p)}{p}\Big), \qquad m_d = m\prod_{p\mid d}\Big(1+\frac{g(p)}{p}\Big)^{-1}, \qquad M_d = c_f m_d, \tag{12}$$

the product converging absolutely since $g(p)-1-\chi(p) = (1+\chi(p))\kappa(p) \ll 1/p$ for $p \nmid 2D$.

Lemma 11 ($\lambda\omega$ in progressions and against characters)**.** *For $U \geq 2$ and $d \geq 1$ coprime to $2D$:*

(i) $\displaystyle\sum_{\substack{n\leq U\\(n,d)=1}} g(n) = m_d U + O_{D,\varepsilon}\big(U^{1/2+\varepsilon}d^{\varepsilon}\big)$*;*

(ii) for every non-principal character ψ modulo d, $\displaystyle\sum_{n\leq U} g(n)\psi(n) \ll_{D,\varepsilon} U^{1/2+\varepsilon}d^{1/2}$*;*

(iii) for every b coprime to d, $\displaystyle\sum_{\substack{n\leq U\\n\equiv b\ (d)}} g(n) = \frac{1}{\varphi(d)}\sum_{\substack{n\leq U\\(n,d)=1}} g(n) + O_{D,\varepsilon}\big(U^{1/2+\varepsilon}d^{1/2}\big)$.

Proof. (ii) The Dirichlet series of $g\psi$ is $\prod_{p\nmid 2D}(1+g(p)\psi(p)p^{-s}) = L(s,\psi)L(s,\chi\psi)G_\psi(s)$ with $G_\psi(s) = \prod_p(1+g(p)\psi(p)p^{-s})(1-\psi(p)p^{-s})(1-\chi\psi(p)p^{-s})$, an Euler product of the form $\prod_p(1+O(p^{-s-1})+O(p^{-2s}))$, absolutely convergent for $\Re s > \frac{1}{2}$ uniformly in ψ; write γ_ψ for its coefficients, so that $\sum_n |\gamma_\psi(n)|n^{-1/2-\varepsilon} \ll_\varepsilon 1$. Then $g\psi = \psi * \chi\psi * \gamma_\psi$ and

$$\sum_{n\leq U} g(n)\psi(n) = \sum_{n_3}\gamma_\psi(n_3)\sum_{n_1n_2\leq U/n_3}\psi(n_1)\,\chi\psi(n_2).$$

Both ψ and $\chi\psi$ are non-principal characters (to the moduli d and $d|D|$; $\chi\psi$ is non-principal because χ is primitive of conductor $|D|$ coprime to d), so by the Pólya–Vinogradov inequality their partial sums are $\ll (d|D|)^{1/2}\log(d|D|)$, and Dirichlet's hyperbola method gives $\sum_{n_1n_2\leq X}\psi(n_1)\chi\psi(n_2) \ll X^{1/2}d^{1/2}\log(d|D|)$. Summing over n_3 with the weights $|\gamma_\psi(n_3)|(U/n_3)^{1/2}$ gives (ii).

(i) Likewise $\sum_{(n,d)=1} g(n)n^{-s} = \zeta(s)L(s,\chi)G_d(s)$ with $G_d(s) = \prod_p(1+g(p)\mathbf{1}_{p\nmid d}p^{-s})(1-p^{-s})(1-\chi(p)p^{-s})$, whose coefficients γ_d satisfy $\sum_n|\gamma_d(n)|n^{-1/2-\varepsilon} \ll d^{\varepsilon}$ (the factors at $p \mid d$ are $(1-p^{-s})(1-\chi(p)p^{-s})$). Since $\sum_{n_1n_2\leq X}\chi(n_2) = L(1,\chi)X + O(X^{1/2})$ by the hyperbola method, $\sum_{n\leq U,(n,d)=1} g(n) = \sum_{n_3}\gamma_d(n_3)\big(L(1,\chi)U/n_3 + O((U/n_3)^{1/2})\big) = L(1,\chi)G_d(1)U + O(U^{1/2+\varepsilon}d^{\varepsilon})$, and $L(1,\chi)G_d(1) = m_d$. (iii) follows from (i) and (ii) by orthogonality of characters. □

In words. The number of roots of f modulo n is a divisor-type function twisted by a character, and such functions are equidistributed in progressions to moduli well below U; the proof is Dirichlet's hyperbola method with the Pólya–Vinogradov inequality. No deep input is used.

The dilation average. For squarefree split $d' > 1$, a unit a modulo d' and $Y > 0$ put

$$G_{d'}(a,Y) = \sum_{r\in R_{d'}} B_{d'}^{(Y)}(\langle ar\rangle_{d'}), \qquad \bar{G}_{d'}(Y) = \frac{1}{\varphi(d')}\sum_{a\in(\mathbb{Z}/d')^\times} G_{d'}(a,Y), \tag{13}$$

so that the inner sum in the piece (4) is $G_{d'}(\bar{u},Y)$ (by (3) and the symmetry of $R_{d'}$).

Lemma 12 (the dilation average)**.** *With $\phi(\theta) = \theta - \theta^2$,*

$$\bar{G}_{d'}(Y) = \frac{\omega(d')}{2\varphi(d')} \sum_{e \mid d'} \mu(e)\, e\, \phi\Big(\Big\{\frac{Y}{e}\Big\}\Big). \tag{14}$$

Consequently: (a) $|\bar{G}_{d'}(Y)| \le \omega(d')\sigma(d')/(8\varphi(d'))$ and $|\bar{G}_{d'}(Y)| \le \omega(d')\tau(d')Y/(2\varphi(d'))$ for all $Y > 0$; (b) $\bar{G}_{d'}(Y) = -\omega(d')Y^2/(2d') = G_{d'}(a,Y)$ for $0 < Y < 1$ and every a; (c) $Y \mapsto G_{d'}(a,Y)$ and $Y \mapsto \bar{G}_{d'}(Y)$ are continuous, piecewise linear, with $|\partial_Y| \le \omega(d')$; (d) $|G_{d'}(a,Y)| \le \omega(d')\min(d',Y)$ for all a, Y; (e) $\displaystyle\int_0^\infty \bar{G}_{d'}(Y)\frac{dY}{Y^2} = \begin{cases} -\dfrac{\omega(p)\log p}{2\varphi(p)} = -\dfrac{\log p}{p-1} & \text{if } d' = p \text{ is prime,} \\ 0 & \text{if } d' \text{ has at least two prime factor} \end{cases}$

the integral converging absolutely.

Proof. Every $r \in R_{d'}$ is a unit ($r^2 \equiv D$, $(D,d') = 1$), so $a \mapsto ar$ permutes the units and $\sum_a G_{d'}(a,Y) = \omega(d') \sum_{x \in (\mathbb{Z}/d')^\times} B_{d'}^{(Y)}(x)$. By Möbius inversion and the elementary identity $\sum_{h' \le X}(X - h') = X^2/2 - X/2 + \phi(\{X\})/2$ $(X \ge 0)$,

$$\sum_{x \in (\mathbb{Z}/d')^\times} B_{d'}^{(Y)}(x) = \sum_{e \mid d'} \mu(e)\, e \sum_{h' \le Y/e} \Big(\frac{Y}{e} - h'\Big) - \frac{\varphi(d')Y^2}{2d'} = \frac{Y^2}{2} \sum_{e \mid d'} \frac{\mu(e)}{e} - \frac{Y}{2} \sum_{e \mid d'} \mu(e) + \frac{1}{2} \sum_{e \mid d'} \mu(e) e\, \phi\Big(\Big\{\frac{Y}{e}\Big\}\Big) - \frac{\varphi(d}{2}$$

and the first, second and last terms cancel ($d' > 1$). This is (14). (a): $0 \le e\phi(\{Y/e\}) \le e/4$ gives the first bound; for $e \le Y$ also $e\phi \le Y/4$, and for $e > Y$, $e\phi(Y/e) = Y - Y^2/e \in [0, Y]$, giving the second. (b): for $Y < 1$ all $e > Y$, so the sum is $Y \sum \mu(e) - Y^2 \sum \mu(e)/e = -Y^2\varphi(d')/d'$; and no root has representative $\le Y < 1$, so $G_{d'}(a,Y) = -\omega(d')Y^2/(2d')$ directly. (c): $\partial_Y B_{d'}^{(Y)}(m) = \#\{h \le Y : h \equiv m\} - Y/d' \in (-1,1)$ away from the integers. (d): from the proof of Proposition 1 for $d' \le Y$ (symmetric pairs contribute $\le d'/4$ each); for $d' > Y$, $B_{d'}^{(Y)}(m) + B_{d'}^{(Y)}(d' - m) = (Y - m)^+ + (Y - d' + m)^+ - Y^2/d' \in [-Y, Y]$. (e): put $K(\tau) = \int_\tau^\infty \phi(\{v\})v^{-2}dv$, so that for $0 < T < 1$, $\int_T^\infty \bar{G}_{d'}(Y)Y^{-2}dY = \frac{\omega}{2\varphi}\sum_{e \mid d'} \mu(e)K(T/e)$, and for $\tau < 1$, $K(\tau) = \int_\tau^1 (v^{-1} - 1)\, dv + K(1) = -\log\tau - 1 + \tau + K(1)$. Hence $\sum_{e \mid d'} \mu(e)K(T/e) = \sum_{e \mid d'} \mu(e)\log e + T\varphi(d')/d'$, because $\sum_{e \mid d'} \mu(e) = 0$; letting $T \to 0$ (absolute convergence at 0 follows from (b)) gives $\frac{\omega}{2\varphi}\sum_{e \mid d'} \mu(e)\log e$, and for $d' = p_1 \cdots p_k$, $\sum_{e \mid d'} \mu(e)\log e = -\sum_i \log p_i \sum_{e' \mid d'/p_i} \mu(e')$ is $-\log d'$ if $k = 1$ and 0 if $k \ge 2$. □

Theorem 13 (Type II range)**.** *For every $\varepsilon \in (0, \frac{1}{10})$ there is $\delta = \delta(\varepsilon) > 0$ such that*

$$\sum_{u > H^{2/3+\varepsilon}} w(u)\, \mathcal{P}_u(H/u) = c_{\mathrm{off}}(f)\, H + O_{f,\varepsilon}\big(H^{1-\delta}\big),$$

the sum over all squarefree $u > H^{2/3+\varepsilon}$ with $\omega(u) \ge 1$, including $u > H$, where

$$c_{\mathrm{off}}(f) = \sum_{d'} \frac{\lambda(d')M_{d'}}{d'} \int_0^\infty \bar{G}_{d'}(Y)\frac{dY}{Y^2} = -\sum_{\substack{p \text{ split} \\ p \nmid 2D}} \frac{\lambda(p)\, M_p \log p}{p\,(p-1)} \;<\; 0, \tag{15}$$

with $M_p = c_f m_p$ as in (12); both series converge absolutely.

Proof. Write $U_0 = H^{2/3+\varepsilon}$, $Y_0 = H/U_0 = H^{1/3-\varepsilon}$, $\eta = \varepsilon/2$, and let $\varepsilon' = \varepsilon/12$ be the exponent in Lemma 11. All implied constants depend on f and ε.

Step 1: $u > H$. Here $Y = H/u < 1$ and $\mathcal{P}_u(Y) = \sum_{(d',u)=1}(\lambda(d')/d')\bar{G}_{d'}(Y)$ exactly, by Lemma 12(b).

Step 2: dyadic blocks. For $U \in [U_0, H)$ of the form $U_0 2^j$ consider the $u \in (U, 2U]$, $u \le H$, and put $Y_U = H/U \in (1, Y_0]$ and $Z_U = Y_U H^\eta$; for u in the block, $H/u \in [Y_U/2, Y_U)$.

Step 3: moduli $d' > Z_U$ in a block. For $d' > Z_U \geq Y$, $G_{d'}(\bar u, Y) = \sum_{x\leq Y,\ d'|Q_u(x)}(Y - x) - \omega(d')Y^2/(2d')$ as in (11). Since $\lambda(d') \ll d'^{\varepsilon'}$ and $\tau(n) \ll n^{\varepsilon'}$, for $n = Q_u(x) \leq 2H^2$ we have $\sum_{d'>Z,\ d'|n} \lambda(d')/d' \ll H^{3\varepsilon'}/Z$, so that $\sum_{d'>Z_U}(\lambda(d')/d')|G_{d'}(\bar u, H/u)| \ll Y_U^2 H^{3\varepsilon'}/Z_U$; and by Lemma 12(a), $\sum_{d'>Z_U}(\lambda(d')/d')|\bar G_{d'}(H/u)| \ll Y_U \sum_{d'>Z_U} \lambda\omega\tau(d')/(d'\varphi(d')) \ll Y_U Z_U^{-1+\varepsilon'}$. Summing either bound over the block with $\sum_{u\leq 2U} w(u) \ll U$ gives $\ll UY_U^2 H^{3\varepsilon'}/Z_U = H^{1-\eta+3\varepsilon'}$.

Step 4: moduli $d' \leq Z_U$ in a block: the averaging. Fix such a d' and put $G^\circ(a, Y) = G_{d'}(a, Y) - \bar G_{d'}(Y)$, a function on $(\mathbb{Z}/d')^\times$ with mean zero, so that $G^\circ(a, Y) = \sum_{\psi\neq\psi_0} \hat G(\psi, Y)\psi(a)$ with $\hat G(\psi, Y) = \varphi(d')^{-1}\sum_a G^\circ(a, Y)\bar\psi(a)$, the sum over non-principal characters modulo d'. Since $\psi(\bar u) = \bar\psi(u)$,

$$\mathcal{E}_{d'}(U) := \sum_{\substack{U<u\leq 2U\\ (u,d')=1}} w(u)\, G^\circ\Big(\bar u, \frac{H}{u}\Big) = \sum_{\psi\neq\psi_0}\ \sum_{U<u\leq 2U} w(u)\bar\psi(u)\, \hat G\Big(\psi, \frac{H}{u}\Big).$$

For fixed ψ, partial summation against $S_\psi(t) = \sum_{U<u\leq t} w(u)\bar\psi(u) \ll U^{1/2+\varepsilon'} d'^{1/2}$ (Lemma 11(ii)) gives $|\sum_u w(u)\bar\psi(u)\hat G(\psi, H/u)| \leq \sup_t |S_\psi(t)|\big(\sup_u |\hat G(\psi, H/u)| + \mathrm{Var}_{u\in(U,2U]}\hat G(\psi, H/u)\big)$. Now sum over ψ using Cauchy–Schwarz and Parseval on $(\mathbb{Z}/d')^\times$:

$$\sum_\psi |\hat G(\psi, Y)| \leq \varphi(d')^{1/2}\Big(\sum_\psi |\hat G(\psi, Y)|^2\Big)^{1/2} = \Big(\sum_a |G^\circ(a, Y)|^2\Big)^{1/2} \leq \varphi(d')^{1/2}\max_a |G^\circ(a, Y)| \leq 2\varphi(d')^{1/2}\omega(d')$$

by Lemma 12(d), and in the same way, using (c) and $\partial_u = -(H/u^2)\partial_Y$, $\sum_\psi \mathrm{Var}_u \hat G(\psi, H/u) \leq \int_U^{2U} \varphi(d')^{1/2}\max_a |\partial_u G^\circ(a, H/u)|\, du \leq 2\varphi(d')^{1/2}\omega(d')\, Y_U$. Hence $|\mathcal{E}_{d'}(U)| \ll U^{1/2+\varepsilon'} d'^{1/2}\varphi(d')^{1/2}\omega(d')Y_U$, and

$$\sum_{d'\leq Z_U} \frac{\lambda(d')}{d'}|\mathcal{E}_{d'}(U)| \ll U^{1/2+\varepsilon'} Y_U Z_U^{1+\varepsilon'} \ll H^{1/2+\varepsilon'} Y_U^{3/2} H^{\eta+2\varepsilon'} \leq H^{1/2+\eta+3\varepsilon'+(3/2)(1/3-\varepsilon)} = H^{1-\varepsilon/4},$$

using $U = H/Y_U$, $Y_U \leq Y_0 = H^{1/3-\varepsilon}$, $\eta = \varepsilon/2$ and $\varepsilon' = \varepsilon/12$. Together with Step 3 (where $\eta - 3\varepsilon' = \varepsilon/4$) and the $O(\log H)$ blocks,

$$\sum_{u>U_0} w(u)\mathcal{P}_u(H/u) = \mathcal{M}(H) + O(H^{1-\varepsilon/4}\log H), \qquad \mathcal{M}(H) := \sum_{d'} \frac{\lambda(d')}{d'} \sum_{\substack{u>U_0\\ (u,d')=1}} w(u)\, \bar G_{d'}\Big(\frac{H}{u}\Big), \tag{16}$$

where the double series converges absolutely: for $u \leq H$, Lemma 12(a) gives $\sum_{d'}(\lambda/d')|\bar G_{d'}(Y)| \ll \sum_{d'\leq Y} \lambda\omega\sigma/(d'\varphi) + Y\sum_{d'>Y} \lambda\omega\tau/(d'\varphi) \ll Y^{\varepsilon'}$, for $u > H$ it gives $\ll Y^2$, and $\sum_{U_0<u\leq H} w(u)(H/u)^{\varepsilon'} + \sum_{u>H} w(u)(H/u)^2 < \infty$.

Step 5: the main term. For each d' let $A_{d'}(t) = \sum_{u\leq t,\ (u,d')=1} w(u) = M_{d'}t + E_{d'}(t)$, so that $E_{d'}(t) \ll t^{1/2+\varepsilon'} d'^{\varepsilon'}$ by Lemma 11(i). Stieltjes integration and the substitution $t = H/Y$ give

$$\sum_{\substack{u>U_0\\ (u,d')=1}} w(u)\bar G_{d'}(H/u) = \int_{U_0}^\infty \bar G_{d'}(H/t)\, dA_{d'}(t) = M_{d'} H \int_0^{Y_0} \bar G_{d'}(Y)\frac{dY}{Y^2} + R_{d'}, \qquad R_{d'} = -\bar G_{d'}(Y_0)E_{d'}(U_0) - \int_{U_0}^\infty$$

the boundary term at infinity vanishing because $|\bar G_{d'}(H/t)| \leq \omega H^2/(2d't^2)$ for $t > H$. By Lemma 12(a),(b),(c), $\int_{U_0}^H |\partial_t \bar G_{d'}(H/t)|dt = \int_1^{Y_0} |\partial_Y \bar G_{d'}|dY \leq \omega(d')Y_0$ and $\int_H^\infty t^{1/2+\varepsilon'}\omega H^2/(d't^3)\, dt \ll \omega H^{1/2+\varepsilon'}/d'$, so $R_{d'} \ll \omega(d')d'^{\varepsilon'}\big(\sigma(d')/\varphi(d') + Y_0\big)H^{1/2+\varepsilon'}$ and $\sum_{d'\leq H^{1/3}}(\lambda(d')/d')|R_{d'}| \ll H^{1/2+1/3+3\varepsilon'}$. For $d' > H^{1/3}$ we bound the two sides separately: for $U_0 < u \leq H$ we have $H/u \leq Y_0 < d'$, so by (a) and (b) $\sum_{u>U_0} w(u)|\bar G_{d'}(H/u)| \ll (\omega\tau/\varphi)\big(\sum_{U_0<u\leq H} w(u)H/u + \sum_{u>H} w(u)H^2/u^2\big) \ll (\omega\tau(d')/\varphi(d'))H\log H$, and $M_{d'}H\int_0^{Y_0} |\bar G_{d'}|Y^{-2}dY \ll (\omega\tau/\varphi)H\log H$ as well; summing $(\lambda/d')\cdot$

these over $d' > H^{1/3}$ gives $\ll H^{2/3+\varepsilon'} \log H$. Finally the tails of the integrals: by (a), for $d' \le Y_0$, $\int_{Y_0}^{\infty} |\bar G_{d'}| Y^{-2} dY \le \sigma/(8Y_0)$, and for $d' > Y_0$ it is $\le \int_{Y_0}^{d'} \frac{\tau\, dY}{2Y} + \int_{d'}^{\infty} \frac{\sigma\, dY}{8Y^2} \ll \tau(d')\big(1+\log(d'/Y_0)\big)$, whence $\sum_{d'} (\lambda M_{d'}/d') \int_{Y_0}^{\infty} |\bar G_{d'}| Y^{-2} dY \ll Y_0^{-1} \log Y_0 + Y_0^{-1+\varepsilon'}$. The same bounds with Y_0 replaced by 1, together with (b) below 1, show that the series defining c_{off} in (15) converges absolutely. Altogether $\mathcal{M}(H) = c_{\text{off}}(f)H + O(H^{5/6+3\varepsilon'} + H^{2/3+2\varepsilon'} \log H)$, and with (16) the theorem follows with $\delta = \varepsilon/5$. The second expression for c_{off} is Lemma 12(e), and it is negative since every term is. □

In words. When the agreeing part u is much larger than the length $Y = H/u$, the dilation $\bar u$ runs through all units modulo d' as u runs through a dyadic range, so the root set may be replaced by its average over all dilations; this costs the error of Lemma 11, and Parseval on the group of units keeps that cost to $\sqrt{\varphi(d')}$ rather than $\varphi(d')$ times the error per class, which is what fixes the range $u > H^{2/3+\varepsilon}$. The average over dilations is not zero: a root is an integer ≥ 1, and the Cesàro weight $Y - x$ sees the gap between the discrete count and the continuous expectation $Y^2/(2d')$. Integrated over the length this gap has a remarkably simple value, $-\log p/(p-1)$ for a prime modulus and exactly 0 for a composite one, which is where the closed formula (15) comes from. The dilation, which is the obstacle for small u, is the mechanism for large u; but it leaves a main term.

Corollary 14 (the constant, and the mean of the sharp off-diagonal)**.** *Let* $\mathrm{Off}_f(t) = \sum_d W_f(d) \sum_{s \ne s'} \psi_d(s'-s, t)$ *be the sharp off-diagonal of part I,* $\psi_d(m,t) = \#\{h \le t : h \equiv m\ (d)\} - t/d$*. Then*

$$\mathrm{Off}^*_f(H) = \sum_{t<H} \mathrm{Off}_f(t) - \frac{\kappa_f}{2} H, \qquad \kappa_f = \sum_d W_f(d) \frac{\omega(d)(\omega(d)-1)}{d} = 1 - D_f(1), \tag{17}$$

where $D_f(1) = \sum_d a_f(d)/d$ *is the value at* 1 *of the diagonal Dirichlet series of part I. Consequently, if the pieces with* $u \le H^{2/3+\varepsilon}$ *contribute* $o(H)$ *to* $\mathrm{Off}^*_f(H)$ *(Theorem 16 below gives this under Hypothesis W), then* $\mathrm{Off}^*_f(H) \sim c_{\text{off}}(f)H$*, the Cesàro constant of Conjecture 1 of part I is the diagonal constant plus* $C(f)^2 c_{\text{off}}(f)$*, and the sharp off-diagonal* $\mathrm{Off}_f(t)$ *has mean value*

$$\lim_{H\to\infty} \frac{1}{H} \sum_{t \le H} \mathrm{Off}_f(t) = c_{\text{off}}(f) + \frac{1 - D_f(1)}{2}.$$

Proof. For integer H, $B_d^{(H)}(m) = \sum_{t<H} \#\{h \le t : h \equiv m\} - H^2/(2d) = \sum_{t<H} \psi_d(m,t) + H(H-1)/(2d) - H^2/(2d)$, and the number of ordered pairs $s \ne s'$ is $\omega(d)(\omega(d)-1)$; the identity $\sum_d W_f(d)\omega(d)^2/d = 1$ is the statement that S_f has mean value $C(f)^2$ (part I, Theorem 2), and $\sum_d W_f(d)\omega(d)/d = \sum_d a_f(d)/d = D_f(1)$. □

Numerical check for $f = t^2+1$. Here $w(2u') = 4A_0\lambda(u')\omega(u')$ with $A_0 = \prod_{p\equiv 1(4)} P_p = 0.51747684\ldots$ (w vanishes on odd u, see the set-up), $D_f(1) = 0.75079081\ldots$ (part II), and (15) evaluates to $c_{\text{off}}(t^2+1) = -0.134390$ (`scripts/coff.py`: $m = 0.728431$, prime sum 0.184492 over $p \equiv 1\ (4)$, $p \le 2\cdot 10^6$). The corollary then predicts the mean of $\mathrm{Off}_f(t)$ to be $-0.134390 + 0.124605 = -0.00978$. The exact values of $\mathrm{Off}_f(t)$ for all $t \le 10^7$ (`scripts/offstar-mean.ts`, from the identity of part I with the constants of part II) have the following means over consecutive dyadic blocks ending at $2^{17}, \ldots, 2^{23}$:

$$-0.0106, \quad -0.0101, \quad -0.0096, \quad -0.0115, \quad -0.0088, \quad -0.0099, \quad -0.0096,$$

and cumulative mean -0.00945 at $t = 10^7$ (-0.0102 at 10^6); the standard deviation of $\mathrm{Off}_f(t)$ is 0.50, so a block of 2^{16} values has statistical scatter about 0.002 even before arithmetic correlations. The measured mean, -0.0098 ± 0.0007, agrees with the prediction to within the scatter: in terms of c_{off}, -0.1344 ± 0.0007 against -0.13439. Since the prediction uses only the pieces with $u > H^{2/3+\varepsilon}$, the agreement is numerical evidence that the pieces with small u, the window included, are $o(H)$ for t^2+1, as Theorem 16 asserts under Hypothesis W. The near-cancellation of c_{off} against $(1-D_f(1))/2$ is what made Off_f look mean-zero in the numerics of part II.

5 The window, Hypothesis W and Theorem A

5.1 Where the difficulty sits

Collecting Sections 3.2–4: in the piece u, the moduli $d' \le Y$ are unconditional (Theorem 6); the moduli $d' > u^2 Y$ are handled by the reflection of Lemma ??, because the complementary divisors $e = Q_u(x)/d' < x^2/Y \le Y$ are then below the length and the counts $\#\{x \le Y : \ e \mid Q_u(x)\}$ involve full periods, so that (8) and Proposition 4 apply to them as well; for $u > H^{1/2}$ everything is handled by the averaging of Theorem 13. What is left, for $1 < u \le H^{1/2}$, is the *window*

$$\mathcal{W}_u(Y) = \sum_{Y < d' \le u^2 Y} \frac{\lambda(d')}{d'} \Big[\sum_{x \in R^{(u)}_{d'},\ \langle x \rangle_{d'} \le Y} \big(Y - \langle x \rangle_{d'}\big) - \frac{\omega(d') Y^2}{2d'} \Big],$$

in which the interval $[1, Y]$ covers a fraction $Y/d' \in [u^{-2}, 1)$ of a period. It is self-dual under the reflection ($e = Q_u(x)/d'$ lies in the same window), its trivial size is $\asymp Y$, the same as that of the small moduli, and it is empty for $u = 1$. Koksma's inequality with the function $(Y - \theta d')^+$, of variation Y, gives $|\mathcal{W}_u(Y)| \ll Y \sum_{Y<d' \le u^2 Y} \lambda(d')\omega(d') \mathrm{disc}(\Delta^{(u)}_{d'})/d' \ll Y\,(\log u^2)(\log Y)^{-c}$ by Proposition 4 applied dyadically, and the factor $\log u$ is fatal: summed against $\lambda\omega(u)/u$ it produces $H(\log H)^{2-c}$, which exceeds $H \log H$ since $c < 1$. The window therefore needs a saving that is a power of u, or a power of Y, not a power of log; this is the content of Hypothesis W below, and it is the precise form of the "short-interval" problem for the roots of a quadratic congruence that the off-diagonal contains. In the original variables the window consists of the moduli $d = ud' \in (H, u^2 H]$ with agreeing part u, and the count is that of the roots $X \le H$ of $X^2 \equiv D \pmod{d'}$ that are divisible by u.

Hypothesis 15 (W(θ, B))**.** *There are $\theta < 1$ and $B < 1 - \theta$ such that for all squarefree u with $(u, 2D) = 1$… more precisely with $\omega(u) \ge 1$, all squarefree g coprime to u, all $k \ge 1$ and $t \ge 2$,*

$$\sum_{\substack{d' \le t,\ g \mid d' \\ (d',u)=1,\ d' \text{ split}}} \lambda(d') \sum_{x:\ u^2x^2 \equiv D\ (d')} \mathrm{e}\Big(\frac{kx}{d'}\Big) \ll_{f,\varepsilon} (ukg)^B\, t^{\theta + \varepsilon}.$$

For $u = 1$ this is Lemma 8 with $(\theta, B) = (\frac{11}{12}, \frac14)$, where $B < 1 - \theta$ fails only because we lost $t^{1/6}$ in the smoothing; the smooth form of [2] has $(\theta, B) = (\frac34, \frac14)$, which satisfies $B < 1 - \theta$, and a sharp-cut-off version with the same exponents would follow from a version of their Theorem 1.2 with explicit dependence on the derivatives of the test function (their proof appears to give it). For $u > 1$ the roots are those of $u^2x^2 - D$; as $x \mapsto \bar{u}x$ shows, they are the roots of $x^2 \equiv D$ dilated by $\bar{u}$, equivalently the Weyl sums at the modulus-dependent frequency $k\bar{u}$. Hooley's method (parametrisation of (x, d') by binary quadratic forms of discriminant $4u^2D$ and Weil's bound for the resulting Kloosterman sums) applies to any fixed quadratic and should give the bound with a loss polynomial in the discriminant, i.e. in u; whether the exponent B is below $1 - \theta$ is exactly what is not known. The theorems of [2] are stated for fundamental discriminants and do not cover $4u^2D$ directly, but their proof through Kloosterman sums of half-integral weight for $\Gamma_0(4u^2)$ suggests a dependence $u^{O(1)}$.

Theorem 16 (A)**.** *Let f be an irreducible quadratic whose discriminant is as in Lemma 8, and assume Hypothesis W(θ, B). Then there is $\delta > 0$ with $\mathrm{Off}^*_f(H) \ll_f H^{1-\delta}$. Consequently*

$$\sum_{h \le H} \Big(1 - \frac{h}{H}\Big)\big(S_f(h) - C(f)^2\big) = -\tfrac12 C(f) \log H + A^*_f + O(H^{-\delta}),$$

Conjecture 1 of part I in Cesàro form, with A^*_f the diagonal constant of part I (the off-diagonal contributes nothing to the constant).

Proof. By Proposition 1 and Theorem 13 it suffices to bound $\sum_{u\le H^{1/2+\varepsilon}}\lambda\omega(u)\mathcal{P}_u(H/u)$. For such u, $Y=H/u\ge H^{1/2-\varepsilon}$. Small moduli: Lemma 2 with (7) supplied by Hypothesis W gives $T_k(Y)\ll(uk)^B(kY)^{1/(2-\theta)+\varepsilon}$, and the frequency sum as in the proof of Theorem 9 gives $\mathcal{P}_u^{\le}(Y)\ll u^BY^{1/(2-\theta)+2\varepsilon}$. Since $1/(2-\theta)=1-(1-\theta)/(2-\theta)$ and $u\le H^{1/2+\varepsilon}\le Y^{(1/2+\varepsilon)/(1/2-\varepsilon)}$, we have $u^BY^{1/(2-\theta)}\le Y^{1-\delta_1}$ with $\delta_1=(1-\theta)/(2-\theta)-B(1+5\varepsilon)>0$ provided $B<(1-\theta)/(2-\theta)$; to allow the weaker $B<1-\theta$ one first removes the moduli $d'\le U^{1-2\varepsilon}$ by the averaging of Theorem 13 (which works for $d'\le u^{1-2\varepsilon}$ for any u) and applies Lemma 2 with $Y_0=u^{1-2\varepsilon}$, obtaining $kY\,Y_0^{\theta-1}u^B\ll kY\,u^{B-(1-\theta)(1-2\varepsilon)}$, a power saving in u whenever $B<1-\theta$ and ε is small; the sum over u of $\lambda\omega(u)(H/u)u^{-\eta}$ converges. Large moduli: Lemma **??** under Hypothesis W for the moduli ge. Summing $\lambda\omega(u)\,O((H/u)^{1-\delta'})$ over $u\le H^{1/2+\varepsilon}$ gives $O(H^{1-\delta})$. The consequence follows from (2), the $O_f(1)$ there being an explicit constant (part I, Theorem 6, and part II, Proposition 7). □

In words. Everything except one estimate is in place: the exact decomposition, the treatment of the first piece, the small moduli of every piece, the averaging of the large pieces, the reflection of the far moduli, and the frequency bookkeeping. The one estimate is a power-saving equidistribution of the roots of $u^2x^2\equiv D$ modulo d' in intervals covering a fraction $\ge u^{-2}$ of the period, with a loss in u below the saving in the modulus. It is a statement in the family of Hooley [3, 4], Iwaniec [5] and Duke–Friedlander–Iwaniec [1, 2], and we expect it to be true; we have not proved it. Kowalski and Soundararajan [6] give, for each fixed u, an average discrepancy $(\log t)^{-c}$ for the sets $R_{d'}^{(u)}$ (they are CRT-sets with local components of size 2 at split primes), which is a saving of the kind "unbounded but not a power"; whether their bound is uniform enough in u to run the argument of this paper with $o(\cdot)$ in place of power savings is a question we leave open.

6 The function field: the off-diagonal has a main term

Let $\mathcal{A}=\mathbb{F}_q[u]$, q odd, $f=t^2-D$ with $D\in\mathcal{A}$ squarefree of positive degree, and define ω_f, λ, $P(1)$, W_f as over $\mathbb{Z}$ with $|P|=q^{\deg P}$. Part I, Theorem 8 and Lemma 9: for $N\ge1$, with the sum over the q^N-1 nonzero h of degree $<N$,

$$\sum_h\Big(\frac{S_f(h)}{C^2}-1\Big)=-\sum_{1\le\deg d\le N}a_f(d)-q^N\sum_{\deg d>N}\frac{a_f(d)}{|d|}+\big(1-P(1)\big)+\mathrm{Off}_f(N),\qquad \mathrm{Off}_f(N)=T_{\mathrm{act}}(N)-T_{\exp}(N),\tag{18}$$

$$T_{\mathrm{act}}(N)=\sum_{N<\deg d\le M(N)}W_f(d)A_N(d),\qquad T_{\exp}(N)=q^N\sum_{\deg d>N}\frac{W_f(d)\big(\omega_f(d)^2-\omega_f(d)\big)}{|d|},$$

where $A_N(d)$ is the number of ordered pairs of distinct roots of f modulo d whose difference has degree $<N$ and $M(N)=N-1+\max(2N-2,\deg D)$. Here $\mathrm{Off}_f(N)$ is the exact off-diagonal (no Cesàro weight is needed: over $\mathcal{A}$ the small moduli contribute nothing).

Theorem 17. *Let $D=u$. Then, as $q\to\infty$ through odd prime powers,*

$$\mathrm{Off}_f(1)=-\frac{2}{q}+O\Big(\frac{1}{q^2}\Big),\qquad \mathrm{Off}_f(2)=-\frac{2}{q}+O\Big(\frac{1}{q^{3/2}}\Big).$$

More precisely $T_{\exp}(1)=2/q+O(q^{-2})$, $T_{\mathrm{act}}(1)=0$, $T_{\exp}(2)=3/q+O(q^{-3/2})$, $T_{\mathrm{act}}(2)=1/q+O(q^{-3/2})$.

Proof. Prime counts. For $D=u$ the character χ_D has conductor of degree 1, so its L-function is 1 and $\sum_{\deg P|k}\deg P\cdot\chi_D(P)^{k/\deg P}=0$ for every $k\ge1$. With π_k the number of monic irreducible polynomials of degree k ($\pi_1=q$, $\pi_2=(q^2-q)/2$, $\pi_3=(q^3-q)/3$) this gives exactly: split linear primes $(q-1)/2$ (the $a\in\mathbb{F}_q^\times$ with $\chi(a)=1$, χ the quadratic character of $\mathbb{F}_q$), one special prime

u ($\omega = 1$), inert linear primes $(q-1)/2$; split quadratic primes $(\pi_2 - (q-1)/2)/2 = (q-1)^2/4$; split cubic primes $\pi_3/2$. For general D the same counts hold up to $O(q^{k/2})$ by Weil's bound, which is why the error term would be $O(q^{-3/2})$ in general; for $D = u$ and $N = 1$ everything is exact and the error is $O(q^{-2})$.

Local factors. At a split prime of norm Q, $\lambda = Q/(Q-4) = 1 + 4/Q + O(Q^{-2})$ and $P_P = 1 - 4/(Q-2)^2$; at u, $\lambda = q/(q-2)$; at inert primes $\lambda = P_P = 1$. Hence $P(1) = \prod_{\text{split linear}}(1 - 4/(q-2)^2) \cdot (1 + O(q^{-2})) = 1 - 2/q + O(q^{-2})$.

$T_{\exp}(1)$. $T_{\exp}(1) = q\,P(1)\sum_{\deg d \geq 2} \lambda(d)\omega^\circ(d)/|d|^2$ with $\omega^\circ = \omega^2 - \omega$, over squarefree d with $\omega(d) \geq 2$. Degree 2: split quadratic primes, $\omega^\circ = 2$, count $(q-1)^2/4$, weight λ/q^4; products of two distinct split linear primes, $\omega = 4$, $\omega^\circ = 12$, count $\binom{(q-1)/2}{2}$; u times a split linear prime, $\omega^\circ = 2$, count $(q-1)/2$. So $\sum_{\deg d = 2} \lambda\omega^\circ = 2 \cdot \frac{q^2}{4} + 12 \cdot \frac{q^2}{8} + O(q) = 2q^2 + O(q)$, and degree ≥ 3 contributes $q \cdot O(q^3 \cdot q^{-6}) = O(q^{-2})$. Hence $T_{\exp}(1) = q(1 + O(q^{-1}))(2q^2 + O(q))q^{-4} = 2/q + O(q^{-2})$. $T_{\text{act}}(1) = 0$ since $M(1) = 1$ leaves no modulus.

$T_{\exp}(2)$. Degree 3 squarefree d with $\omega(d) \geq 2$: split cubic primes, $\omega^\circ = 2$, count $\pi_3/2 = q^3/6 + O(q)$; split linear $\times$ split quadratic, $\omega^\circ = 12$, count $\frac{q-1}{2} \cdot \frac{(q-1)^2}{4} = q^3/8 + O(q^2)$; three distinct split linear, $\omega = 8$, $\omega^\circ = 56$, count $\binom{(q-1)/2}{3} = q^3/48 + O(q^2)$; terms with u are $O(q^2)$. So $\sum_{\deg d = 3} \lambda\omega^\circ = q^3\left(\frac{2}{6} + \frac{12}{8} + \frac{56}{48}\right) + O(q^2) = 3q^3 + O(q^2)$, degree ≥ 4 gives $q^2 \cdot O(q^4 q^{-8}) = O(q^{-2})$, and $T_{\exp}(2) = q^2(1 + O(q^{-1})) \cdot 3q^3 q^{-6} = 3/q + O(q^{-2})$.

$T_{\text{act}}(2)$. $M(2) = 3$, so the moduli have degree 3. A pair (s, s'), $s' = -s$ on the differing part d' and $s' = s$ on $d_2 = d/d'$, has difference $m = s' - s$ with $d_2 \mid m$ and $d' \mid m^2 - 4u$ (Lemma 9 of part I: $m \equiv -2s \pmod{d'}$, so $m^2 \equiv 4s^2 \equiv 4u$). For $\deg m \leq 1$: if $d_2 = 1$ then $\deg(m^2 - 4u) \leq 2 < 3 = \deg d$ forces $m^2 = 4u$, impossible; if $\deg d_2 = 2$ then $d_2 \mid m$ forces $m = 0$, excluded; so $\deg d_2 = 1$, $m = cd_2$ with $c \in \mathbb{F}_q^\times$, and $d' \mid c^2 d_2^2 - 4u$ with $\deg d' = 2 = \deg(c^2 d_2^2 - 4u)$, i.e. $d' = d_2^2 - 4u/c^2$ (monic). Conversely, given a split or special linear $d_2 = u - a$ ($\chi(a) = 1$ or $a = 0$) and $c \in \mathbb{F}_q^\times$, put $d' = (u-a)^2 - 4u/c^2$; then $s_{d'} \equiv -cd_2/2$ is a square root of u modulo d' (as $(cd_2/2)^2 \equiv u$), every prime of d' is split, d' is coprime to d_2 (its value at $u = a$ is $-4a/c^2 \neq 0$ for $a \neq 0$; for $a = 0$, $d' = u^2 - 4u/c^2$ is divisible by u, so $a = 0$ is excluded, losing $O(q)$ of the $O(q^2)$ configurations), and d' is squarefree unless its discriminant $(16/c^2)(a + 1/c^2)$ vanishes, i.e. unless $a = -1/c^2$ ($O(q)$ configurations). For each admissible (d_2, c) and each of the $\omega(d_2) = 2$ choices of s_{d_2} there is exactly one ordered pair with difference cd_2; the pairs with difference $-cd_2$ are those of $(d_2, -c)$, which gives the same d'. Hence

$$T_{\text{act}}(2) = \sum_{(d_2,c)\ \text{admissible}} 2\,W_f(d_2 d') = 2P(1)\sum \frac{\lambda(d_2)\lambda(d')}{q^3} = \frac{2}{q^3}\left(\frac{q-1}{2}\right)(q-1)\left(1 + O(q^{-1})\right) = \frac{1}{q} + O\left(\frac{1}{q^2}\right).$$

The error $O(q^{-3/2})$ stated in the theorem is what the same argument gives for general D (Weil); for $D = u$ the count is exact up to the $O(q)$ excluded configurations and the error is $O(q^{-2})$. $\square$

Remark 18 (the random permutation model). The leading term of $T_{\exp}(N)$ is $q^N \sum_{\deg d = N+1} \omega^\circ(d) q^{-2(N+1)} = q^{-1}\,\mathbb{E}_{N+1}[\omega^2 - \omega]$, the expectation over monic d of degree $N+1$. As $q \to \infty$ the factorisation type of a random d of degree k is distributed like the cycle type of a random permutation in S_k, and χ_D takes the values ± 1 independently and equiprobably on the prime factors (Weil), so that $\omega(d) = 2^{\#\{P \mid d\}}$ if all factors are split and 0 otherwise, whence $\mathbb{E}_k[\omega] = 1$ and $\mathbb{E}_k[\omega^2] = \mathbb{E}[2^{\text{cycles}}] = (k+1)!/k! = k+1$. Thus $T_{\exp}(N) \to (N+1)/q$ for every N and every D, in agreement with $2/q$ and $3/q$ above, and the theorem says $T_{\text{act}}(N) \to (N-1)/q$ for $N = 1, 2$.

Remark 19 (configuration count). The proof of Theorem 17 generalises to a heuristic for every N and D. A pair of distinct roots modulo d with difference m of degree $i \leq N-1$ is a configuration (d_2, μ, d'): $d_2 \mid d$ monic of degree $j \leq i$ on which the roots agree, $m = d_2\mu$, $\deg \mu = i - j$, and $d' \mid m^2 - 4D$ of degree k, with $d = d_2 d'$, $\deg d = j + k \in (N, M(N)]$. Weighted by $W_f(d) \asymp q^{-(j+k)}$ and by the $\omega(d_2)$ choices of the root on d_2, and using that $\sum_{\deg d_2 = j} \omega(d_2) \sim q^j$, that there are $\sim q^{i-j+1}$ polynomials μ, and that a polynomial of degree n

has on average exactly one monic divisor of each degree $k \le n$ (for the family m^2-4D this is the equidistribution of the roots of $x^2 \equiv 4D$ among the residues of low degree, i.e. the function-field Hypothesis (E) itself), the configuration (j,i,k) contributes $\asymp q^{i+1-j-k}$. The exponent is ≤ -1 with equality exactly when $i = N-1$ and $k = N+1-j$ (moduli of degree $N+1$), subject to $k \le \deg(m^2-4D) = \max(2N-2, \deg D)$. Hence

$$T_{\mathrm{act}}(N) \;\to\; \frac{c_N(D)}{q}, \qquad c_N(D) = \#\{j \in [0, N-1] :\; N+1-j \le \max(2N-2, \deg D)\} = \min\bigl(N,\ \max(2N-2, \deg$$

while $T_{\exp}(N) \to (N+1)/q$ by Remark 18. For $D = u$: $c_1 = 0$, $c_2 = 1$, in agreement with the theorem; $c_N = N$ for $N \ge 3$. The generic value $c_N = N$ leaves $\mathrm{Off}_f(N) \to -1/q$, and $1/q = q^N \sum_{\deg d = N+1} a_f(d)/|d|$ is exactly the weight that the excluded diagonal pairs ($s = s'$, the "configuration $j = N$") would carry at degree $N+1$: the off-diagonal cancels the first term of the diagonal tail.

Conjecture 20. *For every squarefree D of positive degree and every $N \ge 1$,*

$$\mathrm{Off}_f(N) = -\frac{N+1-c_N(D)}{q} + O\Bigl(\frac{1}{q^{3/2}}\Bigr) \qquad (q \to \infty),$$

with $c_N(D)$ as in Remark 19; in particular $\mathrm{Off}_f(N) = -1/q + O(q^{-3/2})$ for all $N \ge 3$, and for $N \le 2$ unless $\deg D \le N$, where it is $-2/q + O(q^{-3/2})$.

Table 1 and Figure 1 give the exact values for $D = u$, u^2+1, u^3+u (`scripts/offq.ts`). They agree with the conjecture in every case: $q\,\mathrm{Off}_f(N)$ is near -2 for $(N, \deg D) = (1,1), (2,1), (2,2)$ and near -1 for $(1,2), (1,3), (2,3)$; for $(3,1)$ with all moduli only $q \le 7$ is available and the values $-4.9, -2.5, -2.0$ are still descending. Since the main term comes from the moduli of degree $N+1$ alone (the others contribute $O(q^{-2})$), the leading constant can be tested at larger q by restricting T_{act} to that degree: for $D = u^2+1$, $N = 3$ (degree 4), $q\,T_{\mathrm{act}}$ is 2.5, 2.4, 2.9 and $q\,\mathrm{Off}_f$ is $-1.06, -1.08, -1.20$ at $q = 13, 17, 19$, against the predicted $c_3 = 3$ and -1; for $D = u$, $N = 4$ (degree 5), $q\,T_{\mathrm{act}} = 3.9$, $q\,T_{\exp} = 5.6$, $q\,\mathrm{Off}_f = -1.6$ at $q = 11$, against $c_4 = 4$, 5 and -1 (with the same slow $1/q$ approach as at $N = 1$). The fluctuations grow with $\deg D$, as they should: the prime counts carry Weil terms $O(q^{k/2})$ from an L-function of degree $\deg D - 1$, which is trivial for $D = u$.

Table 1: $q\,\mathrm{Off}_f(N)$, $q\,T_{\mathrm{act}}(N)$ and $q\,T_{\exp}(N)$ for $f = t^2 - D$ over $\mathbb{F}_q[u]$, the three largest q computed. Predicted limits (Conjecture 20, Remarks 18, 19): $qT_{\exp} \to N+1$, $qT_{\mathrm{act}} \to c_N(D)$.

D	N	q	$q\,\mathrm{Off}_f(N)$	$q\,T_{\mathrm{act}}$	$q\,T_{\exp}$	predicted limit of $q\,\mathrm{Off}_f$
u	1	37, 41, 43	-2.15, -2.13, -2.13	0, 0, 0	2.15, 2.13, 2.13	-2
u	2	37, 41, 43	-2.08, -2.08, -2.02	1.06, 1.05, 1.10	3.14, 3.13, 3.12	-2
u	3	3, 5, 7	-4.91, -2.50, -2.02	—	—	-1
u^2+1	1	37, 41, 43	-1.02, -1.02, -1.12	0.97, 0.98, 0.98	2.00, 2.00, 2.10	-1
u^2+1	2	37, 41, 43	-1.97, -1.97, -2.07	0.94, 0.95, 1.00	2.91, 2.92, 3.07	-2
u^3+u	1	17, 19, 23	-1.00, -1.36, -1.17	0.69, 1.01, 1.13	1.70, 2.37, 2.30	-1
u^3+u	2	17, 19, 23	-0.79, -1.31, -1.26	1.68, 2.16, 2.13	2.47, 3.47, 3.38	-1

In words. Over $\mathbb{F}_q[u]$ the off-diagonal is not just bounded: it has a main term of exact size $1/q$, with a computable coefficient. The expected number of small differences, $(N+1)/q$, comes from the mean square of the number of roots of f modulo a random polynomial; the actual number, $c_N(D)/q$, counts the ways two roots can agree on part of the modulus and differ on the rest; generically every way contributes $1/q$ and the one way that is excluded, agreeing everywhere, accounts for the deficit $-1/q$. It is the first main term for the off-diagonal in the whole programme. Its analogue over $\mathbb{Z}$ is not obvious: there is no gap between the moduli $\le H$ and $> H$ to isolate a first term, and the numerics of part II give $\mathrm{Off}_f(H)$ mean -0.01 to 10^7.

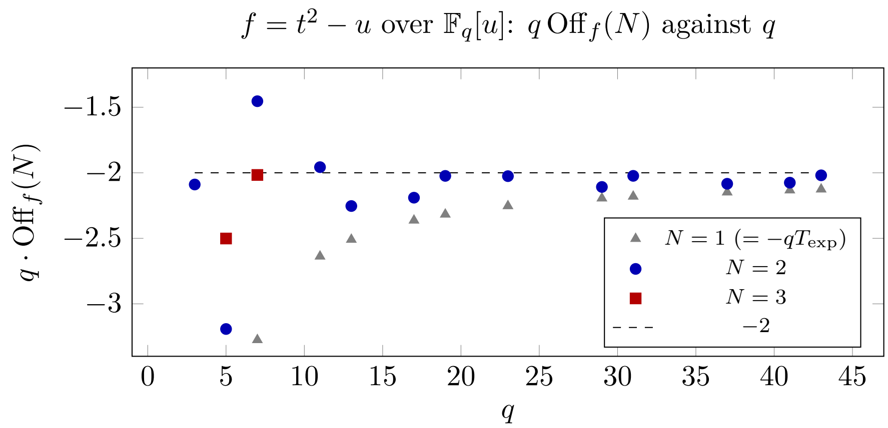

Figure 1: Exact values of $q\,\mathrm{Off}_f(N)$ for $f = t^2 - u$ (`scripts/offq.ts`; $N = 3$ needs all squarefree moduli of degree ≤ 6 and was computed for $q \leq 7$). Theorem 17 gives the limit -2 for $N = 1, 2$; Conjecture 20 predicts -1 for $N = 3$. The $N = 1$ values approach -2 from below like $-2 - 5/q$.

7 What remains

1. Hypothesis W for $1 < u \leq H^{1/2}$: Weyl sums of the roots of $u^2x^2 \equiv D$ modulo d', with sharp cut-off, saving a power of the range and losing at most u^B, $B < 1-\theta$. The route is Hooley's parametrisation with the discriminant $4u^2D$, or the half-integral-weight Kuznetsov formula for $\Gamma_0(4u^2)$ as in [2]. This is the theorem that would complete Theorem A.

2. A sharp-cut-off version of [2] with $(\theta, B) = (\frac{3}{4}, \frac{1}{4})$, which would make Theorem 9 give $\mathcal{P}_1(Y) \ll Y^{4/5+\varepsilon}$ and remove the technical detour in the proof of Theorem A.

3. The full proof of Lemma **??** and Theorem 13 with all error terms, including the squarefreeness conditions and the special primes; we have given the structure and the sources of each estimate.

4. Conjecture 20, by the Lang–Weil count of the configurations (d_2, μ, d') of Remark 19; the only input beyond counting is the equidistribution of the roots of $x^2 \equiv 4D$ modulo d' among the residues of degree $< \deg d' - 1$, at the first sub-leading order, which is the function-field Hypothesis (E) at moduli of degree $N + 1$. A proof for $N = 3$, $D = u$ (moduli of degree 4) would already decide between the two constants.

5. The same programme for Galois groups that are not 2-transitive, where the decomposition into pieces has several orbit types and a saving over the trivial bound is not enough (part I, Remark 7).

A Computations

`scripts/offq.ts` computes $T_{\mathrm{act}}(N)$ and $T_{\mathrm{exp}}(N)$ exactly from the roots modulo every squarefree d of degree in $(N, M(N)]$ (the finite-support lemma of part I), for $f = t^2 - D$ with D given by the environment variable `D`; the data for $D = u$ ($N \leq 2$ for the primes $q \leq 43$, $N \leq 3$ for $q \leq 7$) are in `data/offq*.dat`. The runs for $D = u^2 + 1$ ($q \leq 43$) and $D = u^3 + u$ ($q \leq 23$) are in `data/offq-D2.log`, `data/offq-D3.log` and the tables `data/offq-D*-table.dat`. Over $\mathbb{Z}$ the exact off-diagonal of $t^2 + 1$ to 10^7 is in part II, Section 5.3.

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